2.1 Complex Functions

INTRODUCTION One of the most important concepts in mathematics is that of a function. You may recall from previous courses that a function is a certain kind of correspondence between two sets; more specifically:

> *A **function** f from a set A to a set B is a rule of correspondence that assigns to each element in A one and only one element in B.*

We often think of a function as a rule or a machine that accepts *inputs* from the set A and returns *outputs* in the set B. In elementary calculus we studied functions whose inputs and outputs were real numbers. Such functions are called **real-valued functions of a real variable**. In this section we begin our study of functions whose inputs and outputs are complex numbers. Naturally, we call these functions **complex functions of a complex variable**, or **complex functions** for short. As we will see, many interesting and useful complex functions are simply generalizations of well-known functions from calculus.

A review of function terminology can be found in Section 2.1 of the student study guide.

Function Suppose that f is a function from the set A to the set B. If f assigns to the element a in A the element b in B, then we say that b is the **image** of a under f, or the **value** of f at a, and we write $b = f(a)$. The set A—the set of inputs—is called the **domain** of f and the set of images in B—the set of outputs—is called the **range** of f. We denote the domain and range of a function f by $\text{Dom}(f)$ and $\text{Range}(f)$, respectively. As an example, consider the "squaring" function $f(x) = x^2$ defined for the real variable x. Since any real number can be squared, $\text{Dom}(f)$ is the set $\mathbf{R}$ of all real numbers. The range of f consists of all real numbers x^2 where x is a real number. Of course, $x^2 \geq 0$ for all real x, and it is easy to see from the graph of f that the range of f is the set of all *nonnegative* real numbers. Thus, $\text{Range}(f)$ is the interval $[0, \infty)$. The range of f need not be the same as the set B. For instance, f can be viewed as a function from $A = \mathbf{R}$ to $B = \mathbf{R}$, in which case the range is contained in but not equal to B.

As the following definition indicates, a *complex* function is a function whose inputs and outputs are complex numbers.

DEFINITION 2.1.1 Complex Function

A **complex function** is a function f whose domain and range are subsets of the set $\mathbf{C}$ of complex numbers.

Notation used throughout this text.

A complex function is also called a **complex-valued function of a complex variable**. For the most part we will use the usual symbols f, g, and h to denote complex functions. In addition, inputs to a complex function f will typically be denoted by the variable z and outputs by the variable $w = f(z)$. When referring to a complex function we will use three notations interchangeably, for example, $f(z) = z - i$, $w = z - i$, or, simply, the function $z - i$. Throughout this text the notation $w = f(z)$ will always denote a complex function, whereas the notation $y = f(x)$ will be reserved to represent a real-valued function of a real variable x.

EXAMPLE 1 Complex Function

(**a**) The expression $z^2-(2+i)z$ can be evaluated at any complex number z and always yields a single complex number, and so $f(z) = z^2 - (2 + i)z$ defines a complex function. Values of f are found by using the arithmetic operations for complex numbers given in Section 1.1. For instance, at the points $z = i$ and $z = 1 + i$ we have:

$$f(i) = (i)^2 - (2 + i)(i) = -1 - 2i + 1 = -2i$$

and $$f(1 + i) = (1 + i)^2 - (2 + i)(1 + i) = 2i - 1 - 3i = -1 - i.$$

(**b**) The expression $g(z) = z+2\,\mathrm{Re}\,(z)$ also defines a complex function. Some values of g are:

$$g(i) = i + 2\,\mathrm{Re}\,(i) = i + 2(0) = i$$

and $$g(2 - 3i) = 2 - 3i + 2\,\mathrm{Re}\,(2 - 3i) = 2 - 3i + 2(2) = 6 - 3i.$$ ☰

When the domain of a complex function is not explicitly stated, we assume the domain to be the set of all complex numbers z for which $f(z)$ is defined. This set is sometimes referred to as the *natural domain* of f. For example, the functions $f(z) = z^2 - (2 + i)z$ and $g(z) = z + 2\,\mathrm{Re}(z)$ in Example 1 are defined for all complex numbers z, and so, $\mathrm{Dom}(f) = \mathbf{C}$ and $\mathrm{Dom}(g) = \mathbf{C}$. The complex function $h(z) = z/\left(z^2 + 1\right)$ is not defined at $z = i$ and $z = -i$ because the denominator z^2+1 is equal to 0 when $z = \pm i$. Therefore, $\mathrm{Dom}(h)$ is the set of all complex numbers except i and $-i$.

In the section introduction, we defined a real-valued function of a real variable to be a function whose domain and range are subsets of the set **R** of real numbers. Because **R** is a subset of the set **C** of the complex numbers, *every* real-valued function of a real variable is also a complex function. We will soon see that real-valued functions of two real variables x and y play an important role in the study of complex analysis. In order to avoid repeating the cumbersome terminology *real-valued function of a real variable* and *real-valued function of two real variables,* we use the term **real function** from this point on to refer to any type of function studied in a single or multivariable calculus course.

Real and Imaginary Parts of a Complex Function It is often helpful to express the inputs and the outputs of a complex function in terms of their real and imaginary parts. If $w = f(z)$ is a complex function, then the image of a complex number $z = x+iy$ under f is a complex number $w = u+iv$. By simplifying the expression $f(x + iy)$, we can write the real variables u and v in terms of the real variables x and y. For example, by replacing the symbol z with $x + iy$ in the complex function $w = z^2$, we obtain:

$$w = u + iv = (x + iy)^2 = x^2 - y^2 + 2xyi. \tag{1}$$

From (1) the real variables u and v are given by $u = x^2-y^2$ and $v = 2xy$, respectively. This example illustrates that, if $w = u + iv = f(x + iy)$ is a complex function, then both u and v are real functions of the two real variables x and y. That is, by setting $z = x + iy$, we can express any complex function $w = f(z)$ in terms of two real functions as:

$$f(z) = u(x, y) + iv(x, y). \tag{2}$$

The functions $u(x, y)$ and $v(x, y)$ in (2) are called the **real** and **imaginary parts of** f, respectively.

EXAMPLE 2 Real and Imaginary Parts of a Function

Find the real and imaginary parts of the functions: **(a)** $f(z) = z^2 - (2 + i)z$ and **(b)** $g(z) = z + 2\,\mathrm{Re}(z)$.

Solution In each case, we replace the symbol z by $x + iy$, then simplify.

(a) $f(z) = (x + iy)^2 - (2 + i)(x + iy) = x^2 - 2x + y - y^2 + (2xy - x - 2y)i$. So, $u(x, y) = x^2 - 2x + y - y^2$ and $v(x, y) = 2xy - x - 2y$.

(b) Since $g(z) = x + iy + 2\,\mathrm{Re}(x + iy) = 3x + iy$, we have $u(x, y) = 3x$ and $v(x, y) = y$. ☰

A function f can be defined without using the symbol z.

Every complex function is completely determined by the real functions $u(x, y)$ and $v(x, y)$ in (2). Thus, a complex function $w = f(z)$ can be defined by arbitrarily specifying two real functions $u(x, y)$ and $v(x, y)$, even though $w = u + iv$ may not be obtainable through familiar operations performed solely on the symbol z. For example, if we take, say, $u(x, y) = xy^2$ and $v(x, y) = x^2 - 4y^3$, then $f(z) = xy^2 + i(x^2 - 4y^3)$ defines a complex function. In order to find the value of f at the point $z = 3 + 2i$, we substitute $x = 3$ and $y = 2$ into the expression for f to obtain $f(3 + 2i) = 3 \cdot 2^2 + i(3^2 - 4 \cdot 2^3) = 12 - 23i$.

We note in passing that complex functions defined in terms of $u(x, y)$ and $v(x, y)$ can always be expressed, if desired, in terms of operations on the symbols z and $\bar{z}$. See Problem 32 in Exercises 2.1.

Exponential Function In Section 1.6 we informally introduced the complex exponential function e^z. This complex function is an example of one that is defined by specifying its real and imaginary parts.

DEFINITION 2.1.2 Complex Exponential Function

The function e^z defined by:

$$e^z = e^x \cos y + ie^x \sin y \tag{3}$$

is called the **complex exponential function**.

By Definition 2.1.2, the real and imaginary parts of the complex exponential function are $u(x, y) = e^x \cos y$ and $v(x, y) = e^x \sin y$, respectively. Values of the complex exponential function $w = e^z$ are found by expressing the point z as $z = x + iy$ and then substituting the values of x and y in (3). The following example illustrates this procedure.

EXAMPLE 3 Values of the Complex Exponential Function

Find the values of the complex exponential function e^z at the following points.

(a) $z = 0$ **(b)** $z = i$ **(c)** $z = 2 + \pi i$

Solution In each part we substitute $x = \mathrm{Re}(z)$ and $y = \mathrm{Im}(z)$ into (3) and then simplify.

(**a**) For $z = 0$, we have $x = 0$ and $y = 0$, and so $e^0 = e^0 \cos 0 + ie^0 \sin 0$. Since $e^0 = 1$ (for the real exponential function), $\cos 0 = 1$, and $\sin 0 = 0$, $e^0 = e^0 \cos 0 + i \sin 0$ simplifies to $e^0 = 1$.

(**b**) For $z = i$, we have $x = 0$ and $y = 1$, and so:

$$e^i = e^0 \cos 1 + ie^0 \sin 1 = \cos 1 + i \sin 1 \approx 0.5403 + 0.8415i.$$

(**c**) For $z = 2 + \pi i$, we have $x = 2$ and $y = \pi$, and so $e^{2+\pi i} = e^2 \cos \pi + ie^2 \sin \pi$. Since $\cos \pi = -1$ and $\sin \pi = 0$, this simplifies to $e^{2+\pi i} = -e^2$. ☰

Exponential Form of a Complex Number The complex exponential function enables us to express the polar form of a nonzero complex number $z = r(\cos \theta + i \sin \theta)$ in a particularly convenient and compact form:

$$z = re^{i\theta}. \tag{4}$$

We call (4) the **exponential form** of the complex number z. For example, a polar form of the complex number $3i$ is $3\,[\cos(\pi/2) + i \sin(\pi/2)]$, whereas an exponential form of $3i$ is $3e^{i\pi/2}$. Bear in mind that in the exponential form (4) of a complex number, the value of $\theta = \arg(z)$ is not unique. This is similar to the situation with the polar form of a complex number. You are encouraged to verify that $\sqrt{2}e^{i\pi/4}$, $\sqrt{2}e^{i9\pi/4}$, and $\sqrt{2}e^{i17\pi/4}$ are all valid exponential forms of the complex number $1 + i$.

If z is a real number, that is, if $z = x + 0i$, then (3) gives $e^z = e^x \cos 0 + ie^x \sin 0 = e^x$. In other words, the complex exponential function agrees with the usual real exponential function for real z. Many well-known properties of the real exponential function are also satisfied by the complex exponential function. For instance, if z_1 and z_2 are complex numbers, then (3) can be used to show that:

$$e^0 = 1, \tag{5}$$

$$e^{z_1} e^{z_2} = e^{z_1 + z_2}, \tag{6}$$

$$\frac{e^{z_1}}{e^{z_2}} = e^{z_1 - z_2}, \tag{7}$$

$$\left(e^{z_1}\right)^n = e^{nz_1} \text{ for } n = 0, \pm 1, \pm 2, \ldots. \tag{8}$$

Proofs of properties (5)–(8) will be given in Section 4.1 where the complex exponential function is discussed in greater detail.

While the real and complex exponential functions have many similarities, they also have some surprising and important differences. Perhaps the most unexpected difference is:

• *The complex exponential function is periodic.*

In Problem 33 in Exercises 2.1 you are asked to show that the complex exponential function has pure imaginary period $2\pi i$.

Polar Coordinates Up to this point, the real and imaginary parts of a complex function were determined using the Cartesian description $x + iy$ of the complex variable z. It is equally valid, and, oftentimes, more convenient to express the complex variable z using either the polar form $z = r(\cos \theta + i \sin \theta)$ or, equivalently, the exponential form $z = re^{i\theta}$. Given a complex function $w = f(z)$, if we replace the symbol z with $r(\cos \theta + i \sin \theta)$, then we can write this function as:

$$f(z) = u(r, \theta) + iv(r, \theta). \tag{9}$$

We still call the real functions $u(r,\ \theta)$ and $v(r,\ \theta)$ in (9) the real and imaginary parts of f, respectively. For example, replacing z with $r(\cos\theta + i\sin\theta)$ in the function $f(z) = z^2$, yields, by de Moivre's formula,

$$f(z) = (r(\cos\theta + i\sin\theta))^2 = r^2\cos 2\theta + ir^2\sin 2\theta.$$

Thus, using the polar form of z we have shown that the real and imaginary parts of $f(z) = z^2$ are

The functions $u(r, \theta)$ and $v(r, \theta)$ are not the same as the functions $u(x,\ y)$ and $v(x,\ y)$.

$$u(r, \theta) = r^2\cos 2\theta \quad \text{and} \quad v(r, \theta) = r^2\sin 2\theta, \tag{10}$$

respectively. Because we used a polar rather than a Cartesian description of the variable z, the functions u and v in (10) are not the same as the functions u and v in (1) previously computed for the function z^2.

As with Cartesian coordinates, a complex function can be defined by specifying its real and imaginary parts in polar coordinates. The expression $f(z) = r^3\cos\theta + (2r\sin\theta)i$, therefore, defines a complex function. To find the value of this function at, say, the point $z = 2i$, we first express $2i$ in polar form:

$$2i = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right).$$

We then set $r = 2$ and $\theta = \pi/2$ in the expression for f to obtain:

$$f(2i) = (2)^3\cos\frac{\pi}{2} + \left(2\cdot 2\sin\frac{\pi}{2}\right)i = 8\cdot 0 + (4\cdot 1)i = 4i.$$

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) The complex exponential function provides a good example of how complex functions can be similar to and, at the same time, different from their real counterparts. Both the complex and the real exponential function satisfy properties (5)–(8). On the other hand, the complex exponential function is periodic and, from part (c) of Example 3, a value of the complex exponential function can be a negative real number. Neither of these properties are shared by the real exponential function.

(*ii*) In this section we made the important observation that every complex function can be defined in terms of two real functions $u(x,\ y)$ and $v(x,\ y)$ as $f(z) = u(x,\ y) + iv(x,\ y)$. This implies that the study of complex functions is closely related to the study of real multivariable functions of two real variables. The notions of limit, continuity, derivative, and integral of such real functions will be used to develop and aid our understanding of the analogous concepts for complex functions.

(*iii*) On page 48 we discussed that real-valued functions of a real variable and real-valued functions of two real variables can be viewed as special types of complex functions. Other special types of complex functions that we encounter in the study of complex analysis include the following:

Real-valued functions of a complex variable are functions $y = f(z)$ where z is a complex number and y is a real number. The functions $x = \text{Re}(z)$ and $r = |z|$ are both examples of this type of function.

Complex-valued functions of a real variable are functions $w = f(t)$ where t is a real number and w is a complex number. It is customary to express such functions in terms of two real-valued functions of the real variable t, $w(t) = x(t) + iy(t)$. For example, $w(t) = 3t + i\cos t$ is a complex-valued function of a real variable t.

These special types of complex functions will appear in various places throughout the text.

EXERCISES 2.1

Answers to selected odd-numbered problems begin on page ANS-6.

In Problems 1–8, evaluate the given complex function f at the indicated points.

1. $f(z) = z^2\bar{z} - 2i$	**(a)** $2i$	**(b)** $1 + i$	**(c)** $3 - 2i$
2. $f(z) = -z^3 + 2z + \bar{z}$	**(a)** i	**(b)** $2 - i$	**(c)** $1 + 2i$
3. $f(z) = \log_e \lvert z\rvert + i\,\mathrm{Arg}(z)$	**(a)** 1	**(b)** $4i$	**(c)** $1 + i$
4. $f(z) = \lvert z\rvert^2 - 2\mathrm{Re}(iz) + z$	**(a)** $3 - 4i$	**(b)** $2 - i$	**(c)** $1 + 2i$
5. $f(z) = (xy - x^2) + i(3x + y)$	**(a)** $3i$	**(b)** $4 + i$	**(c)** $3 - 5i$
6. $f(z) = e^z$	**(a)** $2 - \pi i$	**(b)** $\frac{\pi}{3}i$	**(c)** $\log_e 2 - \frac{5\pi}{6}i$
7. $f(z) = r + i\cos^2\theta$	**(a)** 3	**(b)** $-2i$	**(c)** $2 - i$
8. $f(z) = r\sin 3\theta + i\cos 2\theta$	**(a)** -2	**(b)** $1 + i$	**(c)** $-5i$

In Problems 9–16, find the real and imaginary parts u and v of the given complex function f as functions of x and y.

9. $f(z) = 6z - 5 + 9i$

10. $f(z) = -3z + 2\bar{z} - i$

11. $f(z) = z^3 - 2z + 6$

12. $f(z) = z^2 + \bar{z}^2$

13. $f(z) = \dfrac{\bar{z}}{z + 1}$

14. $f(z) = z + \dfrac{1}{z}$

15. $f(z) = e^{2z+i}$

16. $f(z) = e^{z^2}$

In Problems 17–22, find the real and imaginary parts u and v of the given complex function f as functions of r and θ.

17. $f(z) = \bar{z}$

18. $f(z) = \lvert z\rvert$

19. $f(z) = z^4$

20. $f(z) = z + \dfrac{1}{z}$

21. $f(z) = e^z$

22. $f(z) = x^2 + y^2 - yi$

In Problems 23–26, find the natural domain of the given complex function f.

23. $f(z) = 2\mathrm{Re}(z) - iz^2$

24. $f(z) = \dfrac{3z + 2i}{z^3 + 4z^2 + z}$

25. $f(z) = \dfrac{iz}{\lvert z - 1\rvert}$

26. $f(z) = \dfrac{iz}{\lvert z\rvert - 1}$

Focus on Concepts

27. Discuss: Do the following expressions define complex functions $f(z)$? Defend your answer.

(a) $\arg(z)$ **(b)** $\mathrm{Arg}(z)$ **(c)** $\cos(\arg(z)) + i\sin(\arg(z))$

(d) $z^{1/2}$ **(e)** $\lvert z\rvert$ **(f)** $\mathrm{Re}(z)$

28. Find the range of each of the following complex functions.

(**a**) $f(z) = \text{Im}(z)$ defined on the closed disk $|z| \le 2$

(**b**) $f(z) = (1+i)\,|z|$ defined on the square $0 \le \text{Re}(z) \le 1, 0 \le \text{Im}(z) \le 1$

(**c**) $f(z) = \bar{z}$ defined on the upper half-plane $\text{Im}(z) > 0$

29. Find the natural domain and the range of each of the following complex functions.

(**a**) $f(z) = \dfrac{z}{|z|}$. [*Hint*: In order to determine the range, consider $|f(z)|$.]

(**b**) $f(z) = 3 + 4i + \dfrac{5z}{|z|}$.

(**c**) $f(z) = \dfrac{z+\bar{z}}{z-\bar{z}}$.

30. Give an example of a complex function whose natural domain consists of all complex numbers except 0, $1+i$, and $1-i$.

31. Determine the natural domain and range of the complex function $f(z) = \cos(x-y) + i\sin(x-y)$.

32. Suppose that $z = x + iy$. Reread Section 1.1 and determine how to express x and y in terms of z and $\bar{z}$. Then write the following functions in terms of the symbols z and $\bar{z}$.

(**a**) $f(z) = x^2 + y^2$ (**b**) $f(z) = x - 2y + 2 + (6x+y)i$

(**c**) $f(z) = x^2 - y^2 - (5xy)i$ (**d**) $f(z) = 3y^2 + \left(3x^2\right)i$

33. In this problem we examine some properties of the complex exponential function.

(**a**) If $z = x + iy$, then show that $|e^z| = e^x$.

(**b**) Are there any complex numbers z with the property that $e^z = 0$? [*Hint*: Use part (a).]

(**c**) Show that $f(z) = e^z$ is a function that is periodic with pure imaginary period $2\pi i$. That is, show that $e^{z+2\pi i} = e^z$ for all complex numbers z.

34. Use (3) to prove that $e^{\bar{z}} = \overline{e^z}$ for all complex z.

35. What can be said about z if $\left|e^{-z}\right| < 1$?

36. Let $f(z) = \dfrac{e^{iz} + e^{-iz}}{2}$.

(**a**) Show that f is periodic with real period 2π.

(**b**) Suppose that z is real. That is, $z = x + 0i$. Use (3) to rewrite $f(x+0i)$. What well-known real function do you get?

37. What is the period of each of the following complex functions?

(**a**) $f(z) = e^{z+\pi}$ (**b**) $f(z) = e^{\pi z}$

(**c**) $f(z) = e^{2iz}$ (**d**) $f(z) = e^{3z+i}$

38. If $f(z)$ is a complex function with pure imaginary period i, then what is the period of the function $g(z) = f(iz-2)$?

Computer Lab Assignments

In Problems 39–42, use a CAS to evaluate the given complex function at the indicated points. In *Mathematica* you can define a function using the syntax **f[z_]:=** *an expression in the variable* z. To evaluate the function at the point $a+bi$, use the syntax **f[a+b*I]**.

39. the function and points in Problem 1.

40. the function and points in Problem 2.

41. the function and points in Problem 3.

42. the function and points in Problem 4.

2.2 Complex Functions as Mappings

INTRODUCTION Recall that if f is a real-valued function of a real variable, then the graph of f is a curve in the Cartesian plane. Graphs are used extensively to investigate properties of *real* functions in elementary courses. However, we'll see that the graph of a *complex* function lies in four-dimensional space, and so we cannot use graphs to study complex functions. In this section we discuss the concept of a *complex mapping*, which was developed by the German mathematician Bernhard Riemann to give a geometric representation of a complex function. The basic idea is this. Every complex function describes a correspondence between points in two copies of the complex plane. Specifically, the point z in the z-plane is associated with the unique point $w = f(z)$ in the w-plane. We use the alternative term **complex mapping** in place of "complex function" when considering the function as this correspondence between points in the z-plane and points in the w-plane. The geometric representation of a complex mapping $w = f(z)$ consists of two figures: the first, a subset S of points in the z-plane, and the second, the set S' of the images of points in S under $w = f(z)$ in the w-plane.

Mappings

A useful tool for the study of real functions in elementary calculus is the graph of the function. Recall that if $y = f(x)$ is a real-valued function of a real variable x, then the graph of f is defined to be the set of all points $(x, f(x))$ in the two-dimensional Cartesian plane. An analogous definition can be made for a complex function. However, if $w = f(z)$ is a complex function, then both z and w lie in a complex plane. It follows that the set of all points $(z, f(z))$ lies in four-dimensional space (two dimensions from the input z and two dimensions from the output w). Of course, a subset of four-dimensional space cannot be easily illustrated. Therefore:

- *We cannot draw the graph of a complex function.*

The concept of a complex mapping provides an alternative way of giving a geometric representation of a complex function. As described in the section introduction, we use the term **complex mapping** to refer to the correspondence determined by a complex function $w = f(z)$ between points in a z-plane and images in a w-plane. If the point z_0 in the z-plane corresponds to the point w_0 in the w-plane, that is, if $w_0 = f(z_0)$, then we say that f **maps** z_0 onto w_0 or, equivalently, that z_0 is **mapped** onto w_0 by f.

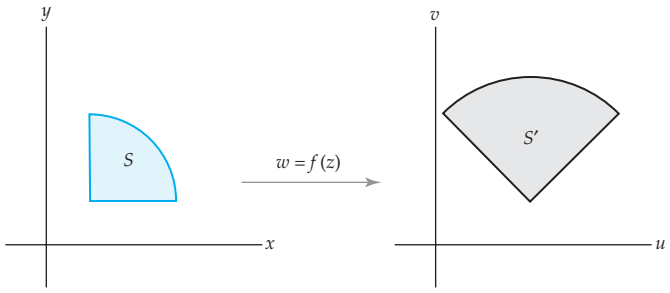
As an example of this type of geometric thinking, consider the real function $f(x) = x + 2$. Rather than representing this function with a line of slope 1 and y-intercept (0, 2), consider how one copy of the real line (the x-line) is mapped onto another copy of the real line (the y-line) by f. Each point on the x-line is mapped onto a point two units to the right on the y-line (0 is mapped onto 2, while 3 is mapped onto 5, and so on). Therefore, the real function $f(x) = x + 2$ can be thought of as a mapping that *translates* each point in the real line two units to the right. You can visualize the action of this mapping by imagining the real line as an infinite rigid rod that is physically moved two units to the right.

In order to create a geometric representation of a complex mapping, we begin with two copies of the complex plane, the z-plane and the w-plane, drawn either side-by-side or one above the other. A complex mapping is represented by drawing a set S of points in the z-plane and the corresponding set of images of the points in S under f in the w-plane. This idea is illustrated in **FIGURE 2.2.1** where a set S in the z-plane is shown in color in Figure 2.2.1(a) and a set labeled S', which represents the set of the images of points in S under $w = f(z)$, is shown in gray in Figure 2.2.1(b). From this point on we will use notation similar to that in Figure 2.2.1 when discussing mappings.

Notation: S'

If $w = f(z)$ is a complex mapping and if S is a set of points in the z-plane, then we call the set of images of the points in S under f the ***image of S under f****, and we denote this set by the symbol S'.**

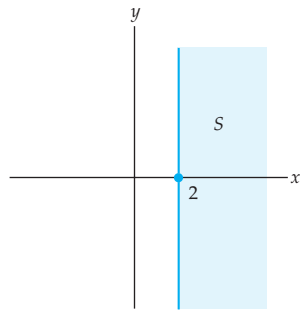
If the set S has additional properties, such as S is a domain or a curve, then we also use symbols such as D and D' or C and C', respectively, to denote the set and its image under a complex mapping. The notation $f(C)$ is also sometimes used to denote the image of a curve C under $w = f(z)$.



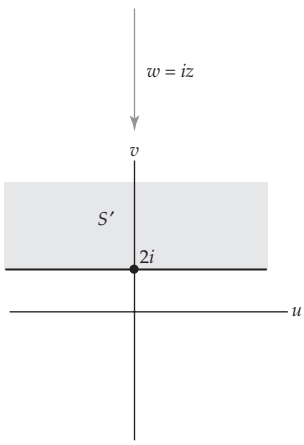
(a) The set S in the z-plane **(b)** The image of S in the w-plane

FIGURE 2.2.1 The image of a set S under a mapping $w = f(z)$

An illustration like Figure 2.2.1 is meant to convey information about the general relationship between an arbitrary point z and its image $w = f(z)$. As such, the set S needs to be chosen with some care. For example, if f is a function whose domain and range are the set of complex numbers $\mathbf{C}$, then choosing $S = \mathbf{C}$ will result in a figure consisting solely of two complex planes. Clearly, such an illustration would give no insight into *how* points in the z-plane are mapped onto points in the w-plane by f.



(a) The half-plane S



(b) The image, S', of the half-plane S

FIGURE 2.2.2 The mapping $w = iz$

EXAMPLE 1 Image of a Half-Plane under $w = iz$

Find the image of the half-plane $\text{Re}(z) \geq 2$ under the complex mapping $w = iz$ and represent the mapping graphically.

Solution Let S be the half-plane consisting of all complex points z with $\text{Re}(z) \geq 2$ shown in color in FIGURE 2.2.2(a). This set can be described by the two simultaneous inequalities,

$$x \geq 2 \quad \text{and} \quad -\infty < y < \infty. \tag{1}$$

In order to describe the image of S, we express the mapping $w = iz$ in terms of its real and imaginary parts u and v; then we use the bounds given by (1) on x and y in the z-plane to determine bounds on u and v in the w-plane. By replacing the symbol z by $x + iy$ in $w = iz$, we obtain $w = i(x + iy) = -y + ix$, and so the real and imaginary parts of $w = iz$ are:

$$u = -y \quad \text{and} \quad v = x. \tag{2}$$

After replacing the variables x and y in (1) using the equations $x = v$ and $y = -u$ from (2), we obtain $v \geq 2$ and $-\infty < u < \infty$. That is, the set S', the image of S under $w = iz$, consists of all points $w = u + iv$ in the w-plane that satisfy the

*The set S is sometimes called the **pre-image** of S' under f.

simultaneous inequalities $v \geq 2$ and $-\infty < u < \infty$. In words, the set S' consists of all points in the half-plane lying on or above the horizontal line $v = 2$. This image can also be described by the single inequality $\text{Im}(w) \geq 2$. In summary, the half-plane $\text{Re}(z) \geq 2$ shown in color in Figure 2.2.2(a) is mapped onto the half-plane $\text{Im}(w) \geq 2$ shown in gray in Figure 2.2.2(b) by the complex mapping $w = iz$.

In Example 1, the set S and its image S' are both half-planes. This might lead you to believe that there is some simple geometric way to visualize the image of other sets in the complex plane under the mapping $w = iz$. (We will see that this is the case in Section 2.3.) For most mappings, however, the relationship between S and S' is more complicated. This is illustrated in the following example.

EXAMPLE 2 Image of a Line under $w = z^2$

Find the image of the vertical line $x = 1$ under the complex mapping $w = z^2$ and represent the mapping graphically.

Solution Let C be the set of points on the vertical line $x = 1$ or, equivalently, the set of points $z = 1 + iy$ with $-\infty < y < \infty$. We proceed as in Example 1. From (1) of Section 2.1, the real and imaginary parts of $w = z^2$ are $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$, respectively. For a point $z = 1 + iy$ in C, we have $u(1, y) = 1 - y^2$ and $v(1, y) = 2y$. This implies that the image of S is the set of points $w = u + iv$ satisfying the simultaneous equations:

$$u = 1 - y^2 \tag{3}$$

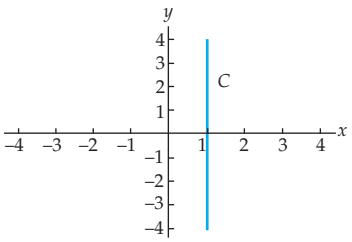
and

$$v = 2y \tag{4}$$

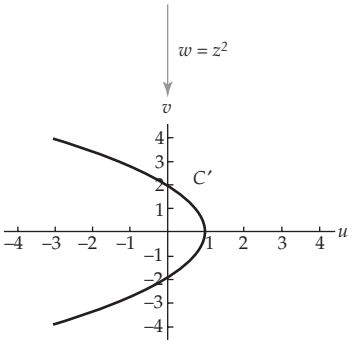
for $-\infty < y < \infty$. Equations (3) and (4) are *parametric equations* in the real parameter y, and they define a curve in the w-plane. We can find a Cartesian equation in u and v for this curve by eliminating the parameter y. In order to do so, we solve (4) for y and then substitute this expression into (3):

$$u = 1 - \left(\frac{v}{2}\right)^2 = 1 - \frac{v^2}{4}. \tag{5}$$

Since y can take on any real value and since $v = 2y$, it follows that v can take on any real value in (5). Consequently, C'—the image of C—is a parabola in the w-plane with vertex at (1,0) and u-intercepts at $(0, \pm 2)$. See FIGURE 2.2.3(b). In conclusion, we have shown that the vertical line $x = 1$ shown in color in Figure 2.2.3(a) is mapped onto the parabola $u = 1 - \frac{1}{4}v^2$ shown in black in Figure 2.2.3(b) by the complex mapping $w = z^2$.



(a) The vertical line $\text{Re}(z) = 1$



(b) The image of C is the parabola $u = 1 - \frac{1}{4}v^2$

FIGURE 2.2.3 The mapping $w = z^2$

In contrast to Figure 2.2.2, the representation of the mapping $w = z^2$ shown in Figure 2.2.3 gives little insight into what the images of other sets in the plane might be. Mapping by this "complex squaring function" will be examined in greater detail in Section 2.4.

Parametric Curves in the Complex Plane

For a simple complex function, the manner in which the complex plane is mapped might be evident after analyzing the image of a single set, but for most functions an understanding of the mapping is obtained only after looking at the images of a variety of sets. We can often gain a good understanding of a complex mapping by analyzing the

images of *curves* (one-dimensional subsets of the complex plane) and this process is facilitated by the use of parametric equations.

A review of parametric curves can be found in Section 2.2 of the student study guide. ▶

If $x = x(t)$ and $y = y(t)$ are real-valued functions of a real variable t, then the set C of all points $(x(t),\ y(t))$, where $a \le t \le b$, is called a **parametric curve**. The equations $x = x(t)$, $y = y(t)$, for $a \le t \le b$ are called **parametric equations** of C. A parametric curve can be regarded as lying in the complex plane by letting x and y represent the real and imaginary parts of a point in the complex plane. In other words, if $x = x(t)$, $y = y(t)$, and $a \le t \le b$ are parametric equations of a curve C in the Cartesian plane, then the equation $z(t) = x(t) + iy(t)$, $a \le t \le b$, is a description of the curve C in the complex plane. For example, consider the parametric equations $x = \cos t$, $y = \sin t$, $0 \le t \le 2\pi$, of a curve C in the xy-plane (the curve C is a circle centered at $(0, 0)$ with radius 1). The equation $z(t) = \cos t + i \sin t$, $0 \le t \le 2\pi$, describes the same curve C in the complex plane. If, say, $t = \pi/2$, then the point $(\cos \frac{\pi}{2}, \sin \frac{\pi}{2}) = (0, 1)$ is on the curve C in the Cartesian plane, while the point $z(\pi/2) = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = i$ represents this point on C in the complex plane. This discussion is summarized in the following definition.

In a parametrization $z(t) = x(t) + iy(t)$, t is a **real** variable. ▶

DEFINITION 2.2.1 Parametric Curves in the Complex Plane

If $x(t)$ and $y(t)$ are real-valued functions of a real variable t, then the set C consisting of all points $z(t) = x(t) + iy(t)$, $a \le t \le b$, is called a **parametric curve** or a **complex parametric curve**. The complex-valued function of the real variable t, $z(t) = x(t) + iy(t)$, is called a **parametrization** of C.

Properties of curves in the Cartesian plane such as continuous, differentiable, smooth, simple, and closed can all be reformulated to be properties of curves in the complex plane. These properties are important in the study of the complex integral and will be discussed in Chapter 5.

Two of the most elementary curves in the plane are lines and circles. Parametrizations of these curves in the complex plane can be derived from parametrizations in the Cartesian plane. It is also relatively easy to find these parametrizations directly by using the geometry of the complex plane. For example, suppose that we wish to find a parametrization of the line in the complex plane containing the points z_0 and z_1. We know from Chapter 1 that $z_1 - z_0$ represents the vector originating at z_0 and terminating at z_1, shown in color in FIGURE 2.2.4. If z is any point on the line containing z_0 and z_1, then inspection of Figure 2.2.4 indicates that the vector $z - z_0$ is a real multiple of the vector $z_1 - z_0$. Therefore, if z is on the line containing z_0 and z_1, then there is a real number t such that $z - z_0 = t\,(z_1 - z_0)$. Solving this equation for z gives a parametrization $z(t) = z_0 + t\,(z_1 - z_0) = z_0\,(1 - t) + z_1 t$, $-\infty < t < \infty$, for the line. Note that if we restrict the parameter t to the interval $[0, 1]$, then the points $z(t)$ range from z_0 to z_1, and this gives a parametrization of the line segment from z_0 to z_1. On the other hand, if we restrict t to the interval $[0, \infty]$, then we obtain a parametrization of the ray emanating from z_0 and containing z_1. These parametrizations are included in the following summary.

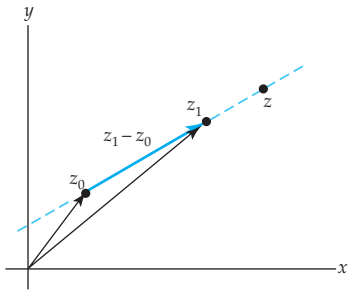


FIGURE 2.2.4 Parametrization of a line

Common Parametric Curves in the Complex Plane

Line

A parametrization of the line containing the points z_0 and z_1 is:

These are not the only parametrizations possible! See Problem 28 in Exercises 2.2. ▶

$$z(t) = z_0(1 - t) + z_1 t, \quad -\infty < t < \infty. \tag{6}$$

Line Segment
A parametrization of the line segment from z_0 to z_1 is:

$$z(t) = z_0(1-t) + z_1 t, \quad 0 \le t \le 1. \tag{7}$$

Ray
A parametrization of the ray emanating from z_0 and containing z_1 is:

$$z(t) = z_0(1-t) + z_1 t, \quad 0 \le t < \infty. \tag{8}$$

Circle
A parametrization of the circle centered at z_0 with radius r is:

$$z(t) = z_0 + r(\cos t + i \sin t), \quad 0 \le t \le 2\pi. \tag{9}$$

In exponential notation, this parametrization is:

$$z(t) = z_0 + re^{it}, \quad 0 \le t \le 2\pi. \tag{10}$$

Parametric curves are important in the study of complex mappings because it is easy to determine a parametrization of the image of a parametric curve. For example, if $w = iz$ and C is the line $x = 2$ given by $z(t) = 2 + it$, $-\infty < t < \infty$, then the value of $f(z) = iz$ at a point on this line is $w = f(2+it) = i(2+it) = -t + 2i$, and so the image of $z(t)$ is $w(t) = -t + 2i$. Put another way, $w(t) = -t + 2i$, $-\infty < t < \infty$, is a parametrization of the image C'. Thus, C' is the line $v = 2$. In summary, we have the following procedure for finding the images of curves under a complex mapping.

Image of a Parametric Curve under a Complex Mapping

If $w = f(z)$ is a complex mapping and if C is a curve parametrized by $z(t)$, $a \le t \le b$, then

$$w(t) = f(z(t)), \quad a \le t \le b \tag{11}$$

is a parametrization of the image, C' of C under $w = f(z)$.

In some instances it is convenient to represent a complex mapping using a single copy of the complex plane. We do so by superimposing the w-plane on top of the z-plane, so that the real and imaginary axes in each copy of the plane coincide. Because such a figure simultaneously represents both the z- and the w-planes, we omit all labels x, y, u, and v from the axes. In the following examples, we represent a complex mapping using a single copy of the complex plane.

EXAMPLE 3 Image of a Parametric Curve

Use (11) to find the image of the line segment from 1 to i under the complex mapping $w = \overline{iz}$.

Solution Let C denote the line segment from 1 to i and let C' denote its image under $f(z) = \overline{iz}$. By identifying $z_0 = 1$ and $z_1 = i$ in (7), we obtain a parametrization $z(t) = 1 - t + it$, $0 \le t \le 1$, of C. The image C' is then given by (11):

$$w(t) = f(z(t)) = \overline{i(1-t+it)} = -i(1-t) - t, \quad 0 \le t \le 1.$$

With the identifications $z_0 = -i$ and $z_1 = -1$ in (7), we see that $w(t)$ is a parametrization of the line segment from $-i$ to -1. Therefore, C' is the line segment from

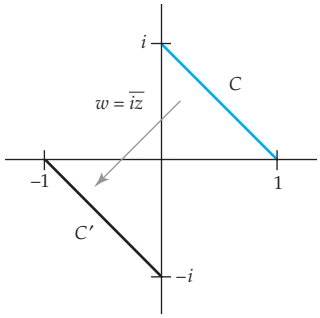


FIGURE 2.2.5 The mapping $w = \overline{iz}$

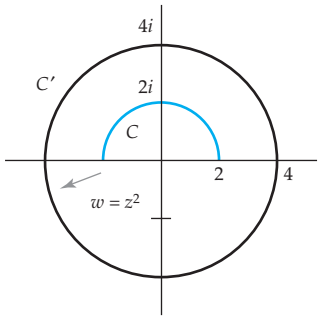


FIGURE 2.2.6 The mapping $w = z^2$

$-i$ to -1. This mapping is depicted in FIGURE 2.2.5 using a single copy of the complex plane. In Figure 2.2.5, the line segment shown in color is mapped onto the line segment shown in black by $w = \overline{iz}$.

EXAMPLE 4 Image of a Parametric Curve

Find the image of the semicircle shown in color in FIGURE 2.2.6 under the complex mapping $w = z^2$.

Solution Let C denote the semicircle shown in Figure 2.2.6 and let C' denote its image under $f(z) = z^2$. By setting $z_0 = 0$ and $r = 2$ in (10) we obtain the following parametrization of C:

$$z(t) = 2e^{it}, \quad 0 \le t \le \pi.$$

Thus, from (11) we have that:

$$w(t) = f(z(t)) = \left(2e^{it}\right)^2 = 4e^{2it}, \quad 0 \le t \le \pi, \tag{12}$$

is a parametrization of C'. If we set $t = \frac{1}{2}s$ in (12), then we obtain a new parametrization of C':

$$W(s) = 4e^{is}, \ 0 \le s \le 2\pi. \tag{13}$$

From (10) with $z_0 = 0$ and $r = 4$, we find that (13) defines a circle centered at 0 with radius 4. Therefore, the image C' is the circle $|w| = 4$. We represent this mapping in Figure 2.2.6 where the semicircle shown in color is mapped onto the circle shown in black by $w = z^2$.

Use of Computers

Computer algebra systems such as *Maple* and *Mathematica* perform standard algebraic operations with complex numbers. This capability combined with the ability to graph a parametric curve makes these systems excellent tools for exploring properties of complex mappings. In *Mathematica*, for example, a complex function can be defined using the command

f[z_] := *an expression in z.*

A complex parametrization can be defined similarly using the command

g[t_] := *an expression in t.*

From (11), it follows that **w[t_] := f[g[t]]** is a parametrization of the image of the curve. This image can be graphed using the parametric plot command:

ParametricPlot[{**Re**[**w**[**t**]], **Im**[**w**[**t**]]}, {**t**, **a**, **b**}]

where **a** and **b** are the upper and lower bounds on **t** respectively. For example, *Mathematica* was used to produce FIGURE 2.2.7, which shows the image of the circle $|z| = 2$ under the complex mapping $w = z^2 + iz - \text{Re}(z)$.

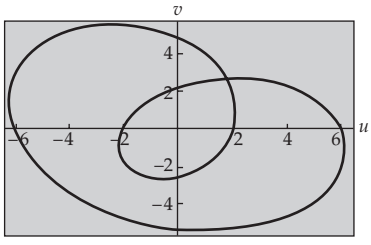


FIGURE 2.2.7 The image of a circle under $w = z^2 + iz - \text{Re}(z)$

Remarks COMPARISON WITH REAL ANALYSIS

(*i*) In this section we introduced an important difference between real and complex analysis, namely, that we cannot graph a complex function. Instead, we represent a complex function with two figures: the first a subset S in the complex plane, and the second, the image S' of the set S under a complex mapping. A complete understanding of a complex mapping is obtained when we understand the relationship between *any* set S and its image S'.

(*ii*) Complex mappings are closely related to parametric curves in the plane. In later sections, we use this relationship to help visualize the notions of limit, continuity, and differentiability of complex functions. Parametric curves will also be of central importance in the study of complex integrals much as they were in the study of real line integrals.

EXERCISES 2.2

Answers to selected odd-numbered problems begin on page ANS-6.

In Problems 1–8, proceed as in Example 1 or Example 2 to find the image S' of the set S under the given complex mapping $w = f(z)$.

1. $f(z) = i\bar{z}$; S is the horizontal line $y = 3$

2. $f(z) = \bar{z}$; S is the line $y = x$

3. $f(z) = 3z$; S is the half-plane $\text{Im}(z) > 2$

4. $f(z) = 3iz$; S is the infinite vertical strip $2 \le \text{Re}(z) < 3$

5. $f(z) = (1 + i)z$; S is the vertical line $x = 2$

6. $f(z) = (1 - i)z$; S is the line $y = 2x + 1$

7. $f(z) = iz + 4$; S is the half-plane $\text{Im}(z) \le 1$

8. $f(z) = iz + \bar{z}$; S is the vertical line $x = 3$

In Problems 9–14, find the image of the given line under the complex mapping $w = z^2$.

9. $y = 1$

10. $x = -3$

11. $x = 0$

12. $y = 0$

13. $y = x$

14. $y = -x$

In Problems 15–20, (**a**) plot the parametric curve C given by $z(t)$ and describe the curve in words, (**b**) find a parametrization of the image, C', of C under the given complex mapping $w = f(z)$, and (**c**) plot C' and describe this curve in words.

15. $z(t) = 2(1 - t) + it, 0 \le t \le 1;\ f(z) = 3z$

16. $z(t) = i(1 - t) + (1 + i)t,\ 0 \le t < \infty;\ f(z) = -z$

17. $z(t) = 1 + 2e^{it},\ 0 \le t \le 2\pi;\ f(z) = z + 1 - i$

18. $z(t) = i + e^{it},\ 0 \le t \le \pi;\ f(z) = (z - i)^2$

19. $z(t) = t,\ 0 \le t \le 2;\ f(z) = e^{i\pi z}$

20. $z(t) = 4e^{it},\ 0 \le t \le \pi,\ f(z) = \text{Re}(z)$

In Problems 21–26, use parametrizations to find the image, C', of the curve C under the given complex mapping $w = f(z)$.

21. $f(z) = z^3$; C is the positive imaginary axis

22. $f(z) = iz$; C is the circle $|z - 1| = 2$

23. $f(z) = 1/z$; C is the circle $|z| = 2$

24. $f(z) = 1/z$; C is the line segment from $1 - i$ to $2 - 2i$

25. $f(z) = z + \bar{z}$; C is the semicircle of the unit circle $|z| = 1$ in the upper half-plane $\text{Im}(z) \geq 0$

26. $f(z) = e^z$; C is the ray emanating from the origin and containing $2 + \sqrt{3}i$

Focus on Concepts

27. In this problem we will find the image of the line $x = 1$ under the complex mapping $w = 1/z$.

(**a**) The line $x = 1$ consists of all points $z = 1 + iy$ where $-\infty < y < \infty$. Find the real and imaginary parts u and v of $f(z) = 1/z$ at a point $z = 1 + iy$ on this line.

(**b**) Show that $\left(u - \frac{1}{2}\right)^2 + v^2 = \frac{1}{4}$ for the functions u and v from part (a).

(**c**) Based on part (b), describe the image of the line $x = 1$ under the complex mapping $w = 1/z$.

(**d**) Is there a point on the line $x = 1$ that maps onto 0? Do you want to alter your description of the image in part (c)?

28. Consider the parametrization $z(t) = i(1 - t) + 3t$, $0 \leq t \leq 1$.

(**a**) Describe in words this parametric curve.

(**b**) What is the difference between the curve in part (a) and the curve defined by the parametrization $z(t) = 3(1 - t) + it$, $0 \leq t \leq 1$?

(**c**) What is the difference between the curve in part (a) and the curve defined by the parametrization $z(t) = \frac{3}{2}t + i\left(1 - \frac{1}{2}t\right)$, $0 \leq t \leq 2$?

(**d**) Find a parametrization of the line segment from $1 + 2i$ to $2 + i$ where the parameter satisfies $0 \leq t \leq 3$.

29. Use parametrizations to find the image of the circle $|z - z_0| = R$ under the mapping $f(z) = iz - 2$. Describe the image in words.

30. Consider the line $y = mx + b$ in the complex plane.

(**a**) Give a parametrization $z(t)$ for the line.

(**b**) Describe in words the image of the line under the complex mapping $w = z + 2 - 3i$.

(**c**) Describe in words the image of the line under the complex mapping $w = 3z$.

31. The complex mapping $w = \bar{z}$ is called **reflection** about the real axis. Explain why.

32. Let $f(z) = az$ where a is a complex constant and $|a| = 1$.

(**a**) Show that $|f(z_1) - f(z_2)| = |z_1 - z_2|$ for all complex numbers z_1 and z_2.

(**b**) Give a geometric interpretation of the result in (a).

(**c**) What does your answer to (b) tell you about the image of a circle under the complex mapping $w = az$.

33. In this problem we investigate the effect of the mapping $w = az$, where a is a complex constant and $a \neq 0$, on angles between rays emanating from the origin.

(**a**) Let C be a ray in the complex plane emanating from the origin. Use parametrizations to show that the image C' of C under $w = az$ is also a ray emanating from the origin.

A **group** is an algebraic structure that occurs in many areas of mathematics. A group is a set G together with a special type of function $*$ from $G \times G$ to G. The function $*$ is called a **binary operation** on G, and it is customary to use the notation $a * b$ instead of $*(a, b)$ to represent a value of $*$. We now give the formal definition of a group. A **group** is a set G together with a binary operation $*$ on G, which satisfies the following three properties:

(i) for all elements a, b, and c in G, $a * (b * c) = (a * b) * c$,

(ii) there exists an element e in G such that $e * a = a * e = a$ for all a in G, and

(iii) for every element a in G there exists an element b in G such that $a * b = b * a = e$. (The element b is called the **inverse** of a in G and is denoted by a^{-1}.)

Let $\text{Isom}_+(\mathbf{E})$ denote the set of all complex functions of the form $f(z) = az + b$ where $|a| = 1$. In the remaining part of this project you are asked to demonstrate that $\text{Isom}_+(\mathbf{E})$ is a group with composition of functions as the binary operation. This group is called the **group of orientation-preserving isometries of the Euclidean plane**.

(**b**) Prove that composition of functions is a binary operation on $\text{Isom}_+(\mathbf{E})$. That is, prove that if f and g are functions in $\text{Isom}_+(\mathbf{E})$, then the function $f \circ g$ defined by $f \circ g(z) = f(g(z))$ is an element in $\text{Isom}_+(\mathbf{E})$.

(**c**) Prove that the set $\text{Isom}_+(\mathbf{E})$ with composition satisfies property (i) of a group.

(**d**) Prove that the set $\text{Isom}_+(\mathbf{E})$ with composition satisfies property (ii) of a group. That is, show that there exists a function e in $\text{Isom}_+(\mathbf{E})$ such that $e \circ f = f \circ e = f$ for all functions f in $\text{Isom}_+(\mathbf{E})$.

(**e**) Prove that the set $\text{Isom}_+(\mathbf{E})$ with composition satisfies property (iii) of a group.

2.4 Special Power Functions

INTRODUCTION A **complex polynomial function** is a function of the form $p(z) = a_n z^n + a_{n-1} z^{n-1} + \ldots + a_1 z + a_0$ where n is a positive integer and a_n, a_{n-1}, $\ldots$, a_1, a_0 are complex constants. In general, a complex polynomial mapping can be quite complicated, but in many special cases the action of the mapping is easily understood. For instance, the complex linear functions studied in Section 2.3 are complex polynomials of degree $n = 1$.

In this section we study complex polynomials of the form $f(z) = z^n$, $n \geq 2$. Unlike the linear mappings studied in the previous section, the mappings $w = z^n$, $n \geq 2$, do not preserve the basic shape of every figure in the complex plane. Associated to the function z^n, $n \geq 2$, we also have the *principal nth root function* $z^{1/n}$. The principal nth root functions are inverse functions of the functions z^n defined on a sufficiently restricted domain. Consequently, complex mappings associated to z^n and $z^{1/n}$ are closely related.

Power Functions

Recall that a real function of the form $f(x) = x^a$, where a is a real constant, is called a power function. We form a complex power function by allowing the input or the exponent a to be a complex number. In other words, a **complex power function** is a function of the form $f(z) = z^\alpha$ where α is a complex constant. If α is an integer, then the power function z^α can be evaluated using the algebraic operations on complex numbers from Chapter 1. For example, $z^2 = z \cdot z$ and $z^{-3} = \dfrac{1}{z \cdot z \cdot z}$. We can also use the formulas for taking roots of complex numbers from Section 1.4 to define power functions with fractional exponents of the form $1/n$. For instance, we *can* define $z^{1/4}$ to be the function that gives the principal fourth root of z. In this section we will restrict our attention to special complex power functions of the form z^n and $z^{1/n}$ where $n \geq 2$ and n is an integer. More complicated

complex power functions such as $z^{\sqrt{2}-i}$ will be discussed in Section 4.2 following the introduction of the complex logarithmic function.

2.4.1 The Power Function z^n

In this subsection we consider complex power functions of the form z^n, $n \geq 2$. It is natural to begin our investigation with the simplest of these functions, the **complex squaring function** z^2.

The Function z^2 Values of the complex power function $f(z) = z^2$ are easily found using complex multiplication. For example, at $z = 2 - i$, we have $f(2-i) = (2-i)^2 = (2-i)\cdot(2-i) = 3-4i$. Understanding the complex mapping $w = z^2$, however, requires a little more work. We begin by expressing this mapping in exponential notation by replacing the symbol z with $re^{i\theta}$:

$$w = z^2 = \left(re^{i\theta}\right)^2 = r^2 e^{i2\theta}. \tag{1}$$

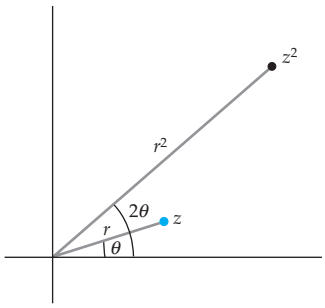
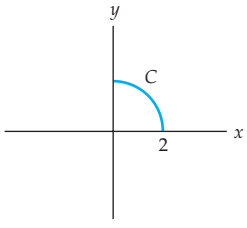


FIGURE 2.4.1 The mapping $w = z^2$

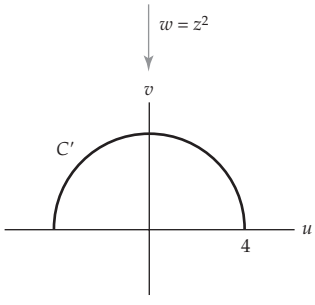
From (1) we see that the modulus r^2 of the point w is the square of the modulus r of the point z, and that the argument 2θ of w is twice the argument θ of z. If we plot both z and w in the same copy of the complex plane, then w is obtained by magnifying z by a factor of r and then by rotating the result through the angle θ about the origin. In FIGURE 2.4.1 we depict the relationship between z and $w = z^2$ when $r > 1$ and $\theta > 0$. If $0 < r < 1$, then z is contracted by a factor of r, and if $\theta < 0$, then the rotation is clockwise.

It is important to note that the magnification or contraction factor and the rotation angle associated to $w = f(z) = z^2$ depend on where the point z is located in the complex plane. For example, since $f(2) = 4$ and $f(i/2) = -\frac{1}{4}$, the point $z = 2$ is magnified by 2 but not rotated, whereas the point $z = i/2$ is contracted by $\frac{1}{2}$ and rotated through $\pi/2$. In general, the squaring function z^2 does not magnify the modulus of points on the unit circle $|z| = 1$ and it does not rotate points on the positive real axis.

The description of the mapping $w = z^2$ in terms of a magnification and rotation can be used to visualize the image of some special sets. The following example illustrates this.



(a) The circular arc of $|z| = 2$



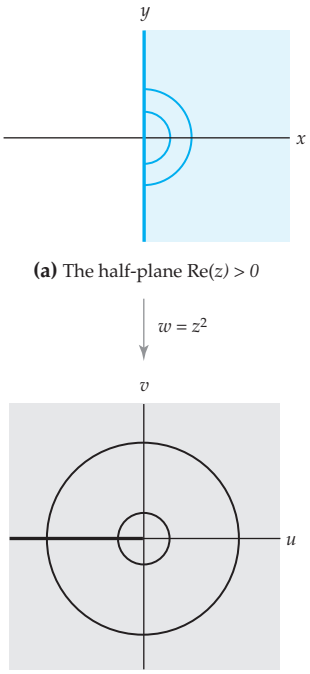
(b) The image of C

FIGURE 2.4.2 The mapping $w = z^2$

EXAMPLE 1 Image of a Circular Arc under $w = z^2$

Find the image of the circular arc defined by $|z| = 2$, $0 \leq \arg(z) \leq \pi/2$, under the mapping $w = z^2$.

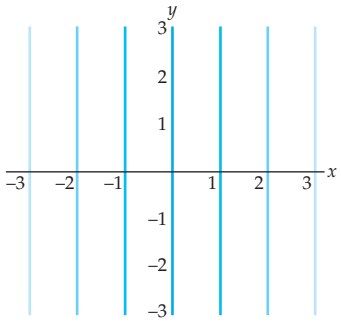
Solution Let C be the circular arc defined by $|z| = 2$, $0 \leq \arg(z) \leq \pi/2$, shown in color in FIGURE 2.4.2(a), and let C' denote the image of C under $w = z^2$. Since each point in C has modulus 2 and since the mapping $w = z^2$ squares the modulus of a point, it follows that each point in C' has modulus $2^2 = 4$. This implies that the image C' must be contained in the circle $|w| = 4$ centered at the origin with radius 4. Since the arguments of the points in C take on every value in the interval $[0, \pi/2]$ and since the mapping $w = z^2$ doubles the argument of a point, it follows that the points in C' have arguments that take on every value in the interval $[2\cdot 0, 2\cdot(\pi/2)] = [0, \pi]$. That is, the set C' is the semicircle defined by $|w| = 4$, $0 \leq \arg(w) \leq \pi$. In conclusion, we have shown that $w = z^2$ maps the circular arc C shown in color in Figure 2.4.2(a) onto the semicircle C' shown in black in Figure 2.4.2(b).



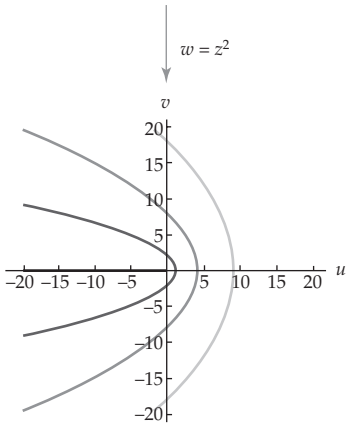
(a) The half-plane Re$(z) > 0$

(b) Image of the half-plane in (a)

FIGURE 2.4.3 The mapping $w = z^2$



(a) Vertical lines in the z-plane



(b) The images of the lines in (a)

FIGURE 2.4.4 The mapping $w = z^2$

An alternative way to find the image in Example 1 would be to use a parametrization. From (10) of Section 2.2, the circular arc C can be parametrized by $z(t) = 2e^{it}$, $0 \le t \le \pi/2$, and by (11) of Section 2.2 its image C' is given by $w(t) = f(z(t)) = 4e^{i2t}$, $0 \le t \le \pi/2$. By replacing the parameter t in $w(t)$ with the new parameter $s = 2t$, we obtain $W(s) = 4e^{is}$, $0 \le s \le \pi$, which is a parametrization of the semicircle $|w| = 4$, $0 \le \arg(w) \le \pi$.

In a similar manner, we find that the squaring function maps a semicircle $|z| = r$, $-\pi/2 \le \arg(z) \le \pi/2$, onto a circle $|w| = r^2$. Since the right half-plane $\text{Re}\,(z) \ge 0$ consists of the collection of semicircles $|z| = r$, $-\pi/2 \le \arg(z) \le \pi/2$, where r takes on every value in the interval $[0, \infty)$, we have that the image of this half-plane consists of the collection of circles $|w| = r^2$ where r takes on any value in $[0, \infty)$. This implies that $w = z^2$ maps the right half-plane $\text{Re}\,(z) \ge 0$ onto the entire complex plane. We illustrate this property in **FIGURE 2.4.3**. Observe that the images of the two semicircles centered at 0 shown in color in Figure 2.4.3(a) are the two circles shown in black in Figure 2.4.3(b). Since $w = z^2$ squares the modulus of a point, the semicircle with smaller radius in Figure 2.4.3(a) is mapped onto the circle with smaller radius in Figure 2.4.3(b), while the semicircle with larger radius in Figure 2.4.3(a) is mapped onto the circle with larger radius in Figure 2.4.3(b). We also see in Figure 2.4.3 that the ray emanating from the origin and containing the point i and the ray emanating from the origin and containing the point $-i$ are both mapped onto the nonpositive real axis. Thus, the imaginary axis shown in color in Figure 2.4.3(a) is mapped onto the set consisting of the point $w = 0$ together with the negative u-axis shown in black in Figure 2.4.3(b).

In order to gain a deeper understanding of the mapping $w = z^2$ we next consider the images of vertical and horizontal lines in the complex plane.

EXAMPLE 2 Image of a Vertical Line under $w = z^2$

Find the image of the vertical line $x = k$ under the mapping $w = z^2$.

Solution In this example it is convenient to work with real and imaginary parts of $w = z^2$ which, from (1) in Section 2.1, are $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$, respectively. Since the vertical line $x = k$ consists of the points $z = k + iy$, $-\infty < y < \infty$, it follows that the image of this line consists of all points $w = u + iv$ where

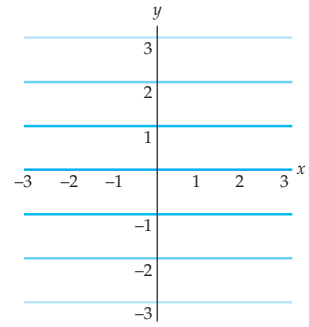
$$u = k^2 - y^2, \quad v = 2ky, \quad -\infty < y < \infty. \tag{2}$$

If $k \neq 0$, then we can eliminate the variable y from (2) by solving the second equation for $y = v/2k$ and then substituting this expression into the remaining equation and inequality. After simplification, this yields:

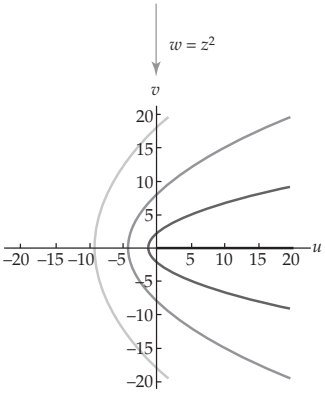
$$u = k^2 - \frac{v^2}{4k^2}, \quad -\infty < v < \infty. \tag{3}$$

Thus, the image of the line $x = k$ (with $k \neq 0$) under $w = z^2$ is the set of points in the w-plane satisfying (3). This image is a parabola that opens in the direction of the negative u-axis, has its vertex at $\left(k^2, 0\right)$, and has v-intercepts at $\left(0, \pm 2k^2\right)$. Notice that the image given by (3) is unchanged if k is replaced by $-k$. This implies that if $k \neq 0$, then the pair of vertical lines $x = k$ and $x = -k$ are both mapped onto the parabola $u = k^2 - v^2/\left(4k^2\right)$ by $w = z^2$.

The action of the mapping $w = z^2$ on vertical lines is depicted in **FIGURE 2.4.4**. The vertical lines $x = k$, $k \neq 0$, shown in color in Figure 2.4.4(a) are mapped onto the parabolas shown in black in Figure 2.4.4(b). In particular, from (3) we have that the lines $x = 3$ and $x = -3$ shown in color in Figure 2.4.4(a) are mapped onto the parabola with vertex at (9, 0) shown in black in Figure 2.4.4(b). In a similar manner,

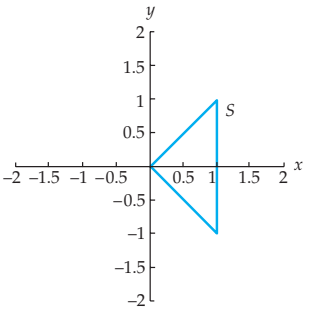


(a) Horizontal lines in the z-plane

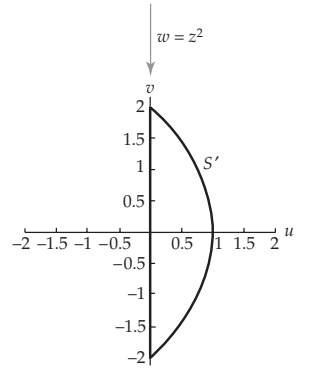


(b) The images of the lines in (a)

FIGURE 2.4.5 The mapping $w = z^2$



(a) A triangle in the z-plane



(b) The image of the triangle in (a)

FIGURE 2.4.6 The mapping $w = z^2$

the lines $x = \pm 2$ are mapped onto the parabola with vertex at $(4, 0)$, and the lines $x = \pm 1$ are mapped onto the parabola with vertex at $(1, 0)$. In the case when $k = 0$, it follows from (2) that the image of the line $x = 0$ (which is the imaginary axis) is given by:

$$u = -y^2, \quad v = 0, \quad -\infty < y < \infty.$$

Therefore, we also have that the imaginary axis is mapped onto the negative real axis by $w = z^2$. See Figure 2.4.4. ☰

With minor modifications, the method of Example 2 can be used to show that a horizontal line $y = k$, $k \neq 0$, is mapped onto the parabola

$$u = \frac{v^2}{4k^2} - k^2 \tag{4}$$

by $w = z^2$. Again we see that the image in (4) is unchanged if k is replaced by $-k$, and so the pair of horizontal lines $y = k$ and $y = -k$, $k \neq 0$, are both mapped by $w = z^2$ onto the parabola given by (4). If $k = 0$, then the horizontal line $y = 0$ (which is the real axis) is mapped onto the positive real axis. Therefore, the horizontal lines $y = k$, $k \neq 0$, shown in color in **FIGURE 2.4.5(a)** are mapped by $w = z^2$ onto the parabolas shown in black in Figure 2.4.5(b). Specifically, the lines $y = \pm 3$ are mapped onto the parabola with vertex at $(-9, 0)$, the lines $y = \pm 2$ are mapped onto the parabola with vertex at $(-4, 0)$, and the lines $y = \pm 1$ are mapped onto the parabola with vertex at $(-1, 0)$.

EXAMPLE 3 Image of a Triangle under $w = z^2$

Find the image of the triangle with vertices 0, $1 + i$, and $1 - i$ under the mapping $w = z^2$.

Solution Let S denote the triangle with vertices at 0, $1+i$, and $1-i$, and let S' denote its image under $w = z^2$. Each of the three sides of S will be treated separately. The side of S containing the vertices 0 and $1+i$ lies on a ray emanating from the origin and making an angle of $\pi/4$ radians with the positive x-axis. By our previous discussion, the image of this segment must lie on a ray making an angle of $2(\pi/4) = \pi/2$ radians with the positive u-axis. Furthermore, since the moduli of the points on the edge containing 0 and $1 + i$ vary from 0 to $\sqrt{2}$, the moduli of the images of these points vary from $0^2 = 0$ to $\left(\sqrt{2}\right)^2 = 2$. Thus, the image of this side is a vertical line segment from 0 to $2i$ contained in the v-axis and shown in black in **FIGURE 2.4.6(b)**. In a similar manner, we find that the image of the side of S containing the vertices 0 and $1 - i$ is a vertical line segment from 0 to $-2i$ contained in the v-axis. See Figure 2.4.6. The remaining side of S contains the vertices $1 - i$ and $1 + i$. This side consists of the set of points $z = 1 + iy$, $-1 \leq y \leq 1$. Because this side is contained in the vertical line $x = 1$, it follows from (2) and (3) of Example 2 that its image is a parabolic segment given by:

$$u = 1 - \frac{v^2}{4}, \quad -2 \leq v \leq 2.$$

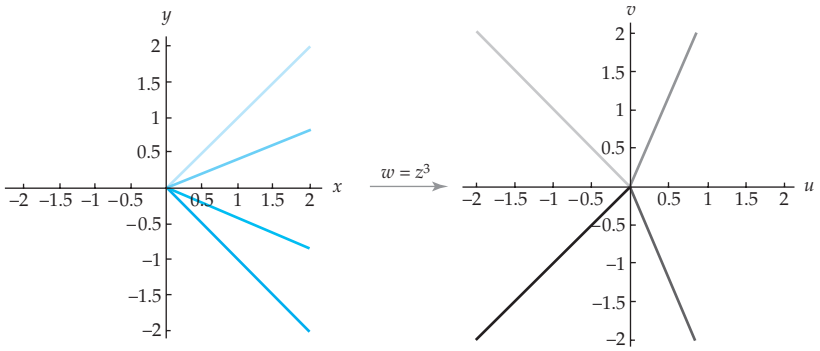
Thus, we have shown that the image of triangle S shown in color in Figure 2.4.6(a) is the figure S' shown in black in Figure 2.4.6(b). ☰

The Function z^n, $n > 2$ An analysis similar to that used for the mapping $w = z^2$ can be applied to the mapping $w = z^n$, $n > 2$. By replacing the symbol z with $re^{i\theta}$ we obtain:

$$w = z^n = r^n e^{in\theta}. \tag{5}$$

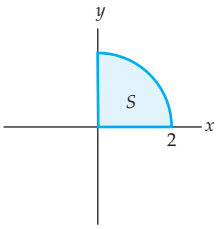
Consequently, if z and $w = z^n$ are plotted in the same copy of the complex plane, then this mapping can be visualized as the process of magnifying or contracting the modulus r of z to the modulus r^n of w, and by rotating z about the origin to increase an argument θ of z to an argument $n\theta$ of w.

We can use this description of $w = z^n$ to show that a ray emanating from the origin and making an angle of ϕ radians with the positive x-axis is mapped onto a ray emanating from the origin and making an angle of $n\phi$ radians with the positive u-axis. This property is illustrated for the mapping $w = z^3$ in FIGURE 2.4.7. Each ray shown in color in Figure 2.4.7(a) is mapped onto a ray shown in black in Figure 2.4.7(b). Since the mapping $w = z^3$ increases the argument of a point by a factor of 3, the ray nearest the x-axis in the first quadrant in Figure 2.4.7(a) is mapped onto the ray in the first quadrant in Figure 2.4.7(b), and the remaining ray in the first quadrant in Figure 2.4.7(a) is mapped onto the ray in the second quadrant in Figure 2.4.7(b). Similarly, the ray nearest the x-axis in the fourth quadrant in Figure 2.4.7(a) is mapped onto the ray in the fourth quadrant in Figure 2.4.7(b), and the remaining ray in the fourth quadrant of Figure 2.4.7(a) is mapped onto the ray in the third quadrant in Figure 2.4.7(b).

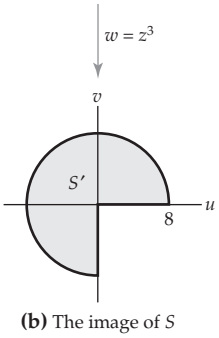


(a) Rays in the z-plane **(b)** Images of the rays in (a)

FIGURE 2.4.7 The mapping $w = z^3$



(a) The set S for Example 4



(b) The image of S

FIGURE 2.4.8 The mapping $w = z^3$

EXAMPLE 4 Image of a Circular Wedge under $w = z^3$

Determine the image of the quarter disk defined by the inequalities $|z| \leq 2$, $0 \leq \arg(z) \leq \pi/2$, under the mapping $w = z^3$.

Solution Let S denote the quarter disk and let S' denote its image under $w = z^3$. Since the moduli of the points in S vary from 0 to 2 and since the mapping $w = z^3$ cubes the modulus of a point, it follows that the moduli of the points in S' vary from $0^3 = 0$ to $2^3 = 8$. In addition, because the arguments of the points in S vary from 0 to $\pi/2$ and because the mapping $w = z^3$ triples the argument of a point, we also have that the arguments of the points in S' vary from 0 to $3\pi/2$. Therefore, S' is the set given by the inequalities $|w| \leq 8, 0 \leq \arg(w) \leq 3\pi/2$, shown in gray in FIGURE 2.4.8(b). In summary, the set S shown in color in Figure 2.4.8(a) is mapped onto the set S' shown in gray in Figure 2.4.8(b) by $w = z^3$. ☰

2.4.2 The Power Function $z^{1/n}$

We now investigate complex power functions of the form $z^{1/n}$ where n is an integer and $n \geq 2$. We begin with the case $n = 2$.

Principal Square Root Function $z^{1/2}$ In (4) of Section 1.4 we saw that the n nth roots of a nonzero complex number $z = r(\cos\theta + i\sin\theta) = re^{i\theta}$ are given by:

$$\sqrt[n]{r}\left[\cos\left(\frac{\theta + 2k\pi}{n}\right) + i\sin\left(\frac{\theta + 2k\pi}{n}\right)\right] = \sqrt[n]{r}e^{i(\theta+2k\pi)/n}$$

where $k = 0,\ 1,\ 2,\ \ldots,\ n-1$. In particular, for $n = 2$, we have that the two square roots of z are:

$$\sqrt{r}\left[\cos\left(\frac{\theta + 2k\pi}{2}\right) + i\sin\left(\frac{\theta + 2k\pi}{2}\right)\right] = \sqrt{r}e^{i(\theta+2k\pi)/2} \tag{6}$$

for $k = 0, 1$. The formula in (6) does not define a function because it assigns *two* complex numbers (one for $k = 0$ and one for $k = 1$) to the complex number z. However, by setting $\theta = \text{Arg}(z)$ and $k = 0$ in (6) we can define a function that assigns to z the unique principal square root. Naturally, this function is called the **principal square root function**.

DEFINITION 2.4.1 Principal Square Root Function

The function $z^{1/2}$ defined by:

$$z^{1/2} = \sqrt{|z|}e^{i\,\text{Arg}(z)/2} \tag{7}$$

is called the **principal square root function**.

If we set $\theta = \text{Arg}(z)$ and replace z with $re^{i\theta}$ in (7), then we obtain an alternative description of the principal square root function for $|z| > 0$:

$$z^{1/2} = \sqrt{r}e^{i\theta/2}, \quad \theta = \text{Arg}(z). \tag{8}$$

Note ▶ You should take note that the symbol $z^{1/2}$ used in Definition 2.4.1 represents something different from the same symbol used in Section 1.4. In (7) we use $z^{1/2}$ to represent the *value* of the principal square root of the complex number z, whereas in Section 1.4 the symbol $z^{1/2}$ was used to represent the *set* of two square roots of the complex number z. This repetition of notation is unfortunate, but widely used. For the most part, the context in which you see the symbol $z^{1/2}$ should make it clear whether we are referring to the principal square root or the set of square roots. In order to avoid confusion we will, at times, also explicitly state "the principal square root function $z^{1/2}$" or "the function $f(z) = z^{1/2}$ given by (7)."

EXAMPLE 5 Values of $z^{1/2}$

Find the values of the principal square root function $z^{1/2}$ at the following points:

(a) $z = 4$ **(b)** $z = -2i$ **(c)** $z = -1 + i$

Solution In each part we use (7) to determine the value of $z^{1/2}$.

(a) For $z = 4$, we have $|z| = |4| = 4$ and $\text{Arg}(z) = \text{Arg}(4) = 0$, and so from (7) we obtain:

$$4^{1/2} = \sqrt{4}e^{i(0/2)} = 2e^{i(0)} = 2.$$

(b) For $z = -2i$, we have $|z| = |-2i| = 2$ and $\text{Arg}(z) = \text{Arg}(-2i) = -\pi/2$, and so from (7) we obtain:

$$(-2i)^{1/2} = \sqrt{2}e^{i(-\pi/2)/2} = \sqrt{2}e^{-i\pi/4} = 1 - i.$$

(c) For $z = -1+i$, we have $|z| = |-1+i| = \sqrt{2}$ and $\text{Arg}(z) = \text{Arg}(-1+i) = 3\pi/4$, and so from (7) we obtain:

$$(-1+i)^{1/2} = \sqrt{(\sqrt{2})}e^{i(3\pi/4)/2} = \sqrt[4]{2}e^{i(3\pi/8)} \approx 0.4551 + 1.0987i. \quad \equiv$$

It is important that we use the *principal argument* when we evaluate the principal square root function of Definition 2.4.1. Using a different choice for the argument of z can give a different function. For example, in Section 1.4 we saw that the two square roots of i are $\frac{1}{2}\sqrt{2} + \frac{1}{2}\sqrt{2}\,i$ and $-\frac{1}{2}\sqrt{2} - \frac{1}{2}\sqrt{2}\,i$. For $z = i$ we have that $|z| = 1$ and $\text{Arg}(z) = \pi/2$. It follows from (7) that:

$$i^{1/2} = \sqrt{1}e^{i(\pi/2)/2} = 1 \cdot e^{i\,\pi/4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i.$$

Therefore, only the first of these two roots of i is the value of the principal square root function. Of course, we can define other "square root" functions. For example, suppose we let θ be the unique value of the argument of z in the interval $\pi < \theta \leq 3\pi$. Then $f(z) = \sqrt{|z|}e^{i\theta/2}$ defines a function for which $f(i) = -\frac{1}{2}\sqrt{2} - \frac{1}{2}\sqrt{2}i$. The function f is not the principal square root function but it is closely related. Since $\pi < \text{Arg}(z) + 2\pi \leq 3\pi$, it follows that $\theta = \text{Arg}(z) + 2\pi$, and so

$$f(z) = \sqrt{|z|}e^{i\theta/2} = \sqrt{|z|}e^{i(\text{Arg}(z)+2\pi)/2} = \sqrt{|z|}e^{i\,\text{Arg}(z)/2}e^{i\pi}.$$

Because $e^{i\pi} = -1$, the foregoing expression simplifies to $f(z) = -\sqrt{|z|}e^{i\,\text{Arg}(z)/2}$. That is, we have shown that the function $f(z) = \sqrt{|z|}e^{i\theta/2}$, $\pi < \theta \leq 3\pi$, is the negative of the principal square root function $z^{1/2}$.

A review of inverse functions can be found in Section 2.4 of the student study guide. ▶

Inverse Functions The principal square root function $z^{1/2}$ given by (7) is *an* inverse function of the squaring function z^2 examined in the first part of this section. Before elaborating on this statement, we need to review some general terminology regarding inverse functions.

A real function must be one-to-one in order to have an inverse function. The same is true for a complex function. The definition of a one-to-one complex function is analogous to that for a real function. Namely, a complex function f is **one-to-one** if each point w in the range of f is the image of a unique point z, called the *pre-image* of w, in the domain of f. That is, f is one-to-one if whenever $f(z_1) = f(z_2)$, then $z_1 = z_2$. Put another way, if $z_1 \neq z_2$, then $f(z_1) \neq f(z_2)$. This says that a one-to-one complex function will not map distinct points in the z-plane onto the same point in the w-plane. For example, the function $f(z) = z^2$ is not one-to-one because $f(i) = f(-i) = -1$. If f is a one-to-one complex function, then for any point w in the range of f there is a unique pre-image in the z-plane, which we denote by $f^{-1}(w)$. This correspondence between a point w and its pre-image $f^{-1}(w)$ defines the inverse function of a one-to-one complex function.

DEFINITION 2.4.2 Inverse Function

If f is a one-to-one complex function with domain A and range B, then the **inverse function of f**, denoted by f^{-1}, is the function with domain B and range A defined by $f^{-1}(z) = w$ if $f(w) = z$.

It follows immediately from Definition 2.4.2 that if a set S is mapped onto a set S' by a one-to-one function f, then f^{-1} maps S' onto S. In other words, the complex mappings f and f^{-1} "undo" each other. It also follows from Definition 2.4.2 that if

f has an inverse function, then $f\left(f^{-1}(z)\right) = z$ and $f^{-1}(f(z)) = z$. That is, the two compositions $f \circ f^{-1}$ and $f^{-1} \circ f$ are the identity function.

EXAMPLE 6 Inverse Function of $f(z) = z + 3i$

Show that the complex function $f(z) = z + 3i$ is one-to-one on the entire complex plane and find a formula for its inverse function.

Solution One way of showing that f is one-to-one is to show that the equality $f(z_1) = f(z_2)$ implies the equality $z_1 = z_2$. For the function $f(z) = z + 3i$, this follows immediately since $z_1 + 3i = z_2 + 3i$ implies that $z_1 = z_2$. As with real functions, the inverse function of f can often be found algebraically by solving the equation $z = f(w)$ for the symbol w. Solving $z = w + 3i$ for w, we obtain $w = z - 3i$ and so $f^{-1}(z) = z - 3i$. ☰

Solve the equation $z = f(w)$ for w to find a formula for $w = f^{-1}(z)$. ▶

From Section 2.3 we know that the mapping $f(z) = z + 3i$ in Example 6 is a translation by $3i$ and that $f^{-1}(z) = z - 3i$ is a translation by $-3i$. Suppose we represent these mappings in a single copy of the complex plane. Since translating a point by $3i$ and then translating the image by $-3i$ moves the original point back onto itself, we see that the mappings f and f^{-1} "undo" each other as expected.

Inverse Functions of z^n, $n > 2$ We now describe how to obtain an inverse function of the power function z^n, $n \geq 2$. This requires some explanation since the function $f(z) = z^n$, $n \geq 2$, is *not* one-to-one. In order to see that this is so, consider the points $z_1 = re^{i\theta}$ and $z_2 = re^{i(\theta+2\pi/n)}$ with $r \neq 0$. Because $n \geq 2$, the points z_1 and z_2 are distinct. That is, $z_1 \neq z_2$. From (5) we have that $f(z_1) = r^n e^{in\theta}$ and $f(z_2) = r^n e^{i(n\theta+2\pi)} = r^n e^{in\theta} e^{i2\pi} = r^n e^{in\theta}$. Therefore, f is not one-to-one since $f(z_1) = f(z_2)$ but $z_1 \neq z_2$. In fact, the n distinct points $z_1 = re^{i\theta}$, $z_2 = re^{i(\theta+2\pi/n)}$, $z_3 = re^{i(\theta+4\pi/n)}$, $\ldots$, $z_n = re^{i(\theta+2(n-1)\pi/n)}$ are all mapped onto the single point $w = r^n e^{in\theta}$ by $f(z) = z^n$. This fact is illustrated for $n = 6$ in FIGURE 2.4.9. The six points $z_1, z_2, \ldots, z_6$ with equal modulus and arguments that differ by $2\pi/6 = \pi/3$ shown in Figure 2.4.9 are all mapped onto the same point by $f(z) = z^6$.

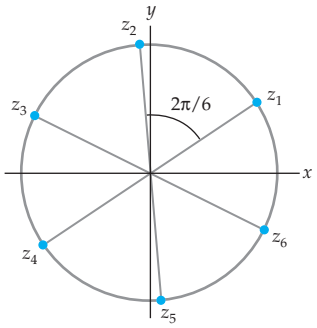
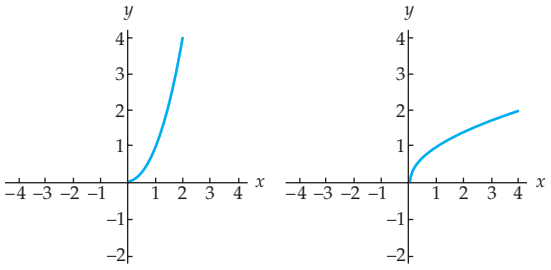


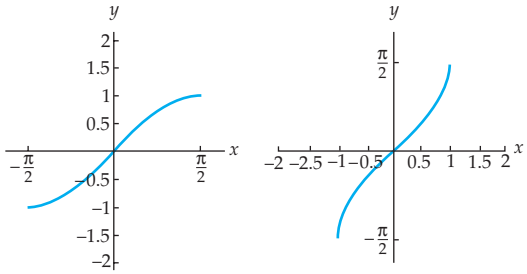
FIGURE 2.4.9 Points mapped onto the same point by $f(z) = z^6$

The preceding discussion appears to imply that the function $f(z) = z^n$, $n \geq 2$, does not have an inverse function because it is not one-to-one. You have encountered this problem before, when defining inverse functions for certain real functions in elementary calculus. For example, the real functions $f(x) = x^2$ and $g(x) = \sin x$ are not one-to-one, yet we still have the inverse functions $f^{-1}(x) = \sqrt{x}$ and $g^{-1}(x) = \arcsin x$. The key to defining these inverse functions is to appropriately restrict the domains of the functions to sets on which the functions are one-to-one. For example, whereas the real function $f(x) = x^2$ defined on the interval $(-\infty, \infty)$ is not one-to-one, the same function defined on the interval $[0, \infty)$ is one-to-one. Similarly, $g(x) = \sin x$ is not one-to-one on the interval $(-\infty, \infty)$, but it is one-to-one on the interval $[-\pi/2,\ \pi/2]$. The function $f^{-1}(x) = \sqrt{x}$ is the inverse function of the function $f(x) = x^2$ defined on the interval $[0,\ \infty)$. Since $\text{Dom}(f) = [0,\ \infty)$ and $\text{Range}(f) = [0,\ \infty)$, the domain and range of $f^{-1}(x) = \sqrt{x}$ are both $[0,\ \infty)$ as well. See FIGURE 2.4.10. In a similar manner, $g^{-1}(x) = \arcsin x$ is the inverse function of the function $g(x) = \sin x$ defined on $[-\pi/2,\ \pi/2]$. The domain and range of g^{-1} are $[-1, 1]$ and $[-\pi/2,\ \pi/2]$, respectively. See FIGURE 2.4.11. We use this same idea for the complex power function z^n, $n \geq 2$. That is, in order to define an inverse function for $f(z) = z^n$, $n \geq 2$, we must restrict the domain of f to a set on which f is a one-to-one function. One such choice of this "restricted domain" when $n = 2$ is found in Example 7.



(a) $f(x) = x^2$ **(b)** $f^{-1}(x) = \sqrt{x}$

FIGURE 2.4.10 $f(x) = x^2$ defined on $[0, \infty)$ and its inverse function



(a) $g(x) = \sin x$ **(b)** $g^{-1}(x) = \arcsin x$

FIGURE 2.4.11 $g(x) = \sin x$ defined on $[-\pi/2, \pi/2]$ and its inverse function

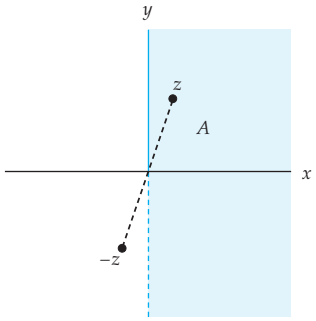


FIGURE 2.4.12 A domain on which $f(z) = z^2$ is one-to-one

EXAMPLE 7 A Restricted Domain for $f(z) = z^2$

Show that $f(z) = z^2$ is a one-to-one function on the set A defined by $-\pi/2 < \arg(z) \leq \pi/2$ and shown in color in FIGURE 2.4.12.

Solution As in Example 6, we show that f is one-to-one by demonstrating that if z_1 and z_2 are in A and if $f(z_1) = f(z_2)$, then $z_1 = z_2$. Suppose that z_1 and z_2 are in A, then we may write $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ with $-\pi/2 < \theta_1 \leq \pi/2$ and $-\pi/2 < \theta_2 \leq \pi/2$. If $f(z_1) = f(z_2)$, then it follows from (1) that:

$$r_1^2 e^{i2\theta_1} = r_2^2 e^{i2\theta_2}. \tag{9}$$

From (9) we conclude that the complex numbers $r_1^2 e^{i2\theta_1}$ and $r_2^2 e^{i2\theta_2}$ have the same modulus and principal argument:

$$r_1^2 = r_2^2 \quad \text{and} \quad \operatorname{Arg}\left(r_1^2 e^{i2\theta_1}\right) = \operatorname{Arg}\left(r_2^2 e^{i2\theta_2}\right). \tag{10}$$

Remember: Arg(z) is in the interval $(-\pi, \pi]$. ▶

Because both r_1 and r_2 are positive, the first equation in (10) implies that $r_1 = r_2$. Moreover, since $-\pi/2 < \theta_1 \leq \pi/2$ and $-\pi/2 < \theta_2 \leq \pi/2$, it follows that $-\pi < 2\theta_1 \leq \pi$, and $-\pi < 2\theta_2 \leq \pi$. This means that $\operatorname{Arg}\left(r_1^2 e^{i2\theta_1}\right) = 2\theta_1$ and $\operatorname{Arg}\left(r_2^2 e^{i2\theta_2}\right) = 2\theta_2$. This fact combined with the second equation in (10) implies that $2\theta_1 = 2\theta_2$, or $\theta_1 = \theta_2$. Therefore, we conclude that $z_1 = z_2$, and this proves that f is a one-to-one function on A. ☰

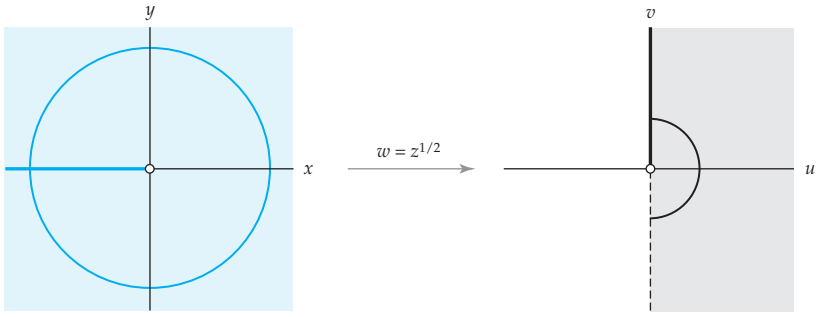
An Inverse of $f(z) = z^2$ In Example 7 we saw that the squaring function z^2 is one-to-one on the set A defined by $-\pi/2 < \arg(z) \leq \pi/2$. It follows from Definition 2.4.2 that this function has a well-defined inverse function f^{-1}. We now proceed to show that this inverse function is the principal square root function $z^{1/2}$ from Definition 2.4.1. In order to do so, let $z = re^{i\theta}$ and $w = \rho e^{i\phi}$ where θ and ϕ are the principal arguments of z and w respectively. Suppose that $w = f^{-1}(z)$. Since

the range of f^{-1} is the domain of f, the principal argument ϕ of w must satisfy:

$$-\frac{\pi}{2} < \phi \leq \frac{\pi}{2}. \tag{11}$$

On the other hand, $f(w) = w^2 = z$, and so w is one of the two square roots of z given by (6). That is, either $w = \sqrt{r}e^{i\theta/2}$ or $w = \sqrt{r}e^{i(\theta+2\pi)/2}$. Because $-\pi < \theta \leq \pi$, we have $-\pi/2 < \theta/2 \leq \pi/2$ and $\pi/2 < (\theta + 2\pi)/2 \leq 3\pi/2$. This means that $\text{Arg}(\sqrt{r}e^{i\theta/2})$ is in the interval $(-\pi/2, \pi/2]$ whereas $\text{Arg}(\sqrt{r}e^{i(\theta+2\pi)/2})$ is either in the interval $(\pi/2, \pi]$ or $(-\pi, -\pi/2]$. Therefore, in order for $\text{Arg}(w) = \phi$ to satisfy (11), we must have $w = \sqrt{r}e^{i\theta/2}$, which is the value of the principal square root function $z^{1/2}$ given by (8).

From Definition 2.4.1, the principal square root function is only defined for $z \neq 0$ (remember that Arg(0) is undefined). That is, the domain of $z^{1/2}$ is the complex plane **C** excluding 0. Since $z^{1/2}$ and z^2 defined on the set $-\pi/2 < \arg(z) \leq \pi/2$ are inverse functions, it follows that the range of $z^{1/2}$ is just the (restricted) domain of z^2. Therefore, the range of $z^{1/2}$ is the set A defined by $-\pi/2 < \arg(w) \leq \pi/2$. These observations regarding the domain and range of the principal square root function are depicted in Figure 2.4.3. The mapping $w = z^{1/2}$ takes the domain shown in color in **FIGURE 2.4.13(a)** onto the set shown in gray in Figure 2.4.13(b). It is important to compare the mappings depicted in Figures 2.4.3 and 2.4.13. Whereas the function z^2 *in general* maps both of the rays $\arg(z) = \pi/2$ and $\arg(z) = -\pi/2$ onto the negative real axis, the function $z^{1/2}$ maps the negative real axis only to the ray $\arg(w) = \pi/2$. This is because the ray $\arg(z) = \pi/2$ is the unique choice of these two rays that lies in the restricted domain for z^2 used to define its inverse function $z^{1/2}$.



(a) Domain of $z^{1/2}$ **(b)** Range of $z^{1/2}$

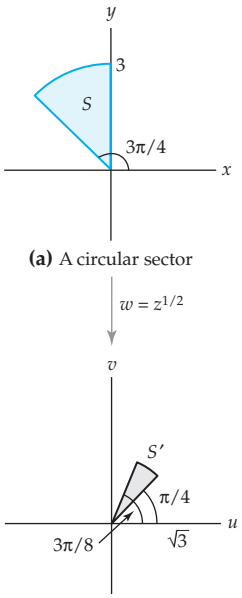
FIGURE 2.4.13 The principal square root function $w = z^{1/2}$

The Mapping $w = z^{1/2}$ As a mapping, the function z^2 squares the modulus of a point and doubles its argument. Because the principal square root function $z^{1/2}$ is an inverse function of z^2, it follows that the mapping $w = z^{1/2}$ takes the square root of the modulus of a point and halves its principal argument. That is, if $w = z^{1/2}$, then we have $|w| = \sqrt{|z|}$ and $\text{Arg}(w) = \frac{1}{2}\text{Arg}(z)$. These relations follow directly from (7) and are helpful in determining the images of sets under $w = z^{1/2}$.

EXAMPLE 8 Image of a Circular Sector under $w = z^{1/2}$

Find the image of the set S defined by $|z| \leq 3$, $\pi/2 \leq \arg(z) \leq 3\pi/4$, under the principal square root function.

Solution Let S' denote the image of S under $w = z^{1/2}$. Since $|z| \leq 3$ for points in S and since $z^{1/2}$ takes the square root of the modulus of a point, we must have that



(a) A circular sector

(b) The image of the set in (a)

FIGURE 2.4.14 The mapping $w = z^{1/2}$

$|w| \leq \sqrt{3}$ for points w in S'. In addition, since $\pi/2 \leq \arg(z) \leq 3\pi/4$ for points in S and since $z^{1/2}$ halves the argument of a point, it follows that $\pi/4 \leq \arg(w) \leq 3\pi/8$ for points w in S'. Therefore, we have shown that the set S shown in color in **FIGURE 2.4.14(a)** is mapped onto the set S' shown in gray in Figure 2.4.14(b) by $w = z^{1/2}$. ☰

Principal *n*th Root Function By modifying the argument given in Example 7 that the function $f(z) = z^2$ is one-to-one on the set defined by $-\pi/2 < \arg(z) \leq \pi/2$, we can show that the complex power function $f(z) = z^n$, $n > 2$, is one-to-one on the set defined by

$$-\frac{\pi}{n} < \arg(z) \leq \frac{\pi}{n}. \tag{12}$$

It is also relatively easy to see that the image of the set defined by (12) under the mapping $w = z^n$ is the entire complex plane **C** excluding $w = 0$. Therefore, there is a well-defined inverse function for f. Analogous to the case $n = 2$, this inverse function of z^n is called the **principal *n*th root function** $z^{1/n}$. The domain of $z^{1/n}$ is the set of all nonzero complex numbers, and the range of $z^{1/n}$ is the set of complex numbers w satisfying $-\pi/n < \arg(w) \leq \pi/n$. A purely algebraic description of the principal nth root function is given by the following formula, which is analogous to (7).

DEFINITION 2.4.3 Principal *n*th Root Functions

For $n \geq 2$, the function $z^{1/n}$ defined by:

$$z^{1/n} = \sqrt[n]{|z|}\, e^{i\,\mathrm{Arg}(z)/n} \tag{13}$$

is called the **principal *n*th root function**.

Notice that the principal square root function $z^{1/2}$ from Definition 2.4.1 is simply a special case of (13) with $n = 2$. Notice also that in Definition 2.4.3 we use the symbol $z^{1/n}$ to represent something different than the same symbol used in Section 1.4. As with the symbol $z^{1/2}$, whether $z^{1/n}$ represents the principal nth root or the set of principal nth roots will be clear from the context or stated explicitly.

By setting $z = re^{i\theta}$ with $\theta = \mathrm{Arg}(z)$ we can also express the principal nth root function as:

$$z^{1/n} = \sqrt[n]{r}\, e^{i\theta/n}, \quad \theta = \mathrm{Arg}(z). \tag{14}$$

EXAMPLE 9 Values of $z^{1/n}$

Find the value of the given principal nth root function $z^{1/n}$ at the given point z.

(a) $z^{1/3}$; $z = i$ **(b)** $z^{1/5}$; $z = 1 - \sqrt{3}i$

Solution In each part we use (13).

(a) For $z = i$, we have $|z| = 1$ and $\mathrm{Arg}(z) = \pi/2$. Substituting these values in (13) with $n = 3$ we obtain:

$$i^{1/3} = \sqrt[3]{1}\, e^{i(\pi/2)/3} = e^{i\pi/6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i.$$

(b) For $z = 1 - \sqrt{3}i$, we have $|z| = 2$ and $\mathrm{Arg}(z) = -\pi/3$. Substituting these values in (13) with $n = 5$ we obtain:

$$\left(1 - \sqrt{3}i\right)^{1/5} = \sqrt[5]{2}\, e^{i(-\pi/3)/5} = \sqrt[5]{2}\, e^{-i(\pi/15)} \approx 1.1236 - 0.2388i. \quad ☰$$

Multiple-Valued Functions In Section 1.4 we saw that a nonzero complex number z has n distinct nth roots in the complex plane. This means that the process of "taking the nth root" of a complex number z *does not* define a complex function because it assigns a set of n complex numbers to the complex number z. We introduced the symbol $z^{1/n}$ in Section 1.4 to represent the set consisting of the n nth roots of z. A similar type of process is that of finding the argument of a complex number z. Because the symbol $\arg(z)$ represents an infinite set of values, it also does not represent a complex function. These types of operations on complex numbers are examples of **multiple-valued functions**. This term often leads to confusion since a multiple-valued function is *not* a function; a function, by definition, must be *single-valued*. However unfortunate, the term *multiple-valued function* is a standard one in complex analysis and so we shall use it from this point on. We will adopt the following functional notation for multiple-valued functions.

Notation: Multiple-Valued Functions

When representing multiple-valued functions with functional notation, we will use uppercase letters such as $F(z) = z^{1/2}$ or $G(z) = \arg(z)$. Lower-case letters such as f and g will be reserved to represent functions.

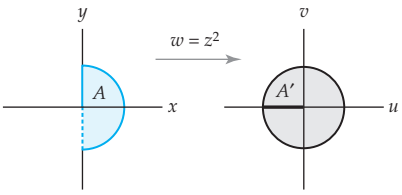
This notation will help avoid confusion associated to the symbols like $z^{1/n}$. For example, we should assume that $g(z) = z^{1/3}$ refers to the principal cube root function defined by (13) with $n = 3$, whereas, $G(z) = z^{1/3}$ represents the multiple-valued function that assigns the three cube roots of z to the value of z. Thus, we have $g(i) = \frac{1}{2}\sqrt{3} + \frac{1}{2}i$ from Example 9(a) and $G(i) = \left\{\frac{1}{2}\sqrt{3} + \frac{1}{2}i, -\frac{1}{2}\sqrt{3} + \frac{1}{2}i, -i\right\}$ from Example 1 of Section 1.4.

Remarks

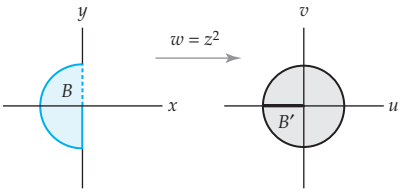
(*i*) In (5) in Exercises 1.4 we defined a rational power of z. One way to define a power function $z^{m/n}$ where m/n is a rational number is as a composition of the principal nth root function and the power function z^m. That is, we can define $z^{m/n} = \left(z^{1/n}\right)^m$. Thus, for $m/n = 2/3$ and $z = 8i$ you should verify that $(8i)^{2/3} = 2 + 2\sqrt{3}i$. Of course, using a root other than the principal nth root gives a possibly different function.

(*ii*) You have undoubtedly noticed that the complex linear mappings studied in Section 2.3 are much easier to visualize than the mappings by complex power functions studied in this section. In part, mappings by complex power functions are more intricate because these functions are not one-to-one. The visualization of a complex mapping that is **multiple-to-one** is difficult and it follows that the multiple-valued functions, which are "inverses" to the multiple-to-one functions, are also hard to visualize. A technique attributed to the mathematician Bernhard Riemann (1826–1866) for visualizing multiple-to-one and multiple-valued functions is to construct a **Riemann surface** for the mapping. Since a rigorous description of Riemann surfaces is beyond the scope of this text, our discussion of these surfaces will be informal.

We begin with a description of a Riemann surface for the complex squaring function $f(z) = z^2$ defined on the closed unit disk $|z| \leq 1$.



(a) A maps onto A'

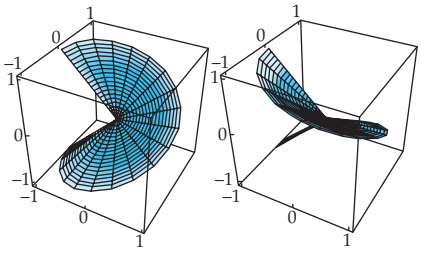


(b) B maps onto B'

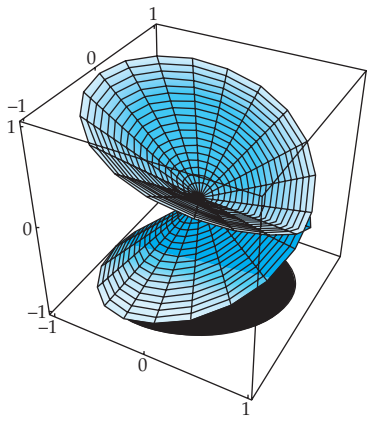
FIGURE 2.4.15 The mapping $w = z^2$



FIGURE 2.4.16 The cut disks A' and B'



(a) The cut disks A' and B' in xyz-space

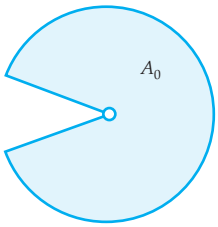


(b) Riemann surface in xyz-space

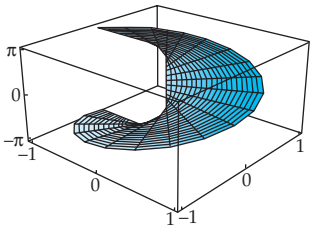
FIGURE 2.4.17 A Riemann surface for $f(z) = z^2$

On page 80 we saw follows from Example 7 that $f(z) = z^2$ is one-to-one on the set A defined by $|z| \leq 1$, $-\pi/2 < \arg(z) \leq \pi/2$. Under the complex mapping $w = z^2$, the set A shown in color in **FIGURE 2.4.15(a)** is mapped onto the closed unit disk $|w| \leq 1$ shown in gray in Figure 2.4.15(a). In a similar manner, we can show that $w = z^2$ is a one-to-one mapping of the set B defined by $|z| \leq 1$, $\pi/2 < \arg(z) \leq 3\pi/2$, onto the closed unit disk $|w| \leq 1$. See Figure 2.4.15(b). Since the unit disk $|z| \leq 1$ is the union of the sets A and B, the image of the disk $|z| \leq 1$ under $w = z^2$ covers the disk $|w| \leq 1$ twice (once by A and once by B). We visualize this "covering" by considering *two* image disks for $w = z^2$. Let A' denote the image of A under f and let B' denote the image of B under f. See Figure 2.4.15. Now imagine that the disks A' and B' have been cut open along the negative real axis as shown in **FIGURE 2.4.16**. The edges shown in black in Figure 2.4.16 of the cut disks A' and B' are the images of edges shown in color of A and B, respectively. Similarly, the dashed edges of A' and B' shown in Figure 2.4.16 are the images of the dashed edges of A and B, respectively. We construct a Riemannn surface for $f(z) = z^2$ by stacking the cut disks A' and B' one atop the other in xyz-space and attaching them by gluing together their edges. The black edge of A' shown in Figure 2.4.16 is glued to the dashed edge of B' shown in Figure 2.4.16, and the dashed edge of A' is glued to the black edge of B'. After attaching in this manner we obtain the Riemann surface shown in **FIGURE 2.4.17**. We assume that this surface is lying directly above the closed unit disk $|w| \leq 1$. Although $w = z^2$ is not a one-to-one mapping of the closed unit disk $|z| \leq 1$ onto the closed unit disk $|w| \leq 1$, it is a one-to-one mapping of the closed unit disk $|z| \leq 1$ onto the Riemann surface that we have constructed. Moreover, the two-to-one covering nature of the mapping $w = z^2$ can be visualized by mapping the disk $|z| \leq 1$ onto the Riemann surface, then projecting the points of the Riemann surface vertically down onto the disk $|w| \leq 1$. In addition, by reversing the order in this procedure, we can also use this Riemann surface to help visualize the multiple-valued function $F(z) = z^{1/2}$.

Another interesting Riemann surface is one for the multiple-valued function $G(z) = \arg(z)$ defined on the punctured disk $0 < |z| \leq 1$. To construct this surface, we take a copy of the punctured disk $0 < |z| \leq 1$ and cut it open along the negative real axis. See **FIGURE 2.4.18(a)**. Call this cut disk A_0 and let it represent the points in the domain written in exponential notation as $re^{i\theta}$ with $-\pi < \theta \leq \pi$. Take another copy of the cut disk, call it A_1, and let it represent the points in the domain written as $re^{i\theta}$ with $\pi < \theta \leq 3\pi$. Similarly, let A_{-1} be a cut disk that represents the points in the domain written as $re^{i\theta}$ with $-3\pi < \theta \leq -\pi$. Continue in this manner to produce an infinite set of cut disks $\ldots$ A_{-2}, A_{-1}, A_0, A_1, A_2 $\ldots$. In general, the cut disk A_n represents points in the domain of G expressed as $re^{i\theta}$ with $(2n-1)\pi < \theta \leq (2n+1)\pi$. Now place each disk A_n in xyz-space so that the point $re^{i\theta}$ with $(2n-1)\pi < \theta \leq (2n+1)\pi$ lies at height θ directly above the point $re^{i\theta}$ in the xy-plane. The cut disk A_0 placed in xyz-space is shown in Figure 2.4.18(b). The collection of all the cut disks in xyz-space forms the Riemann surface for the multiple-valued function $G(z) = \arg(z)$ shown in **FIGURE 2.4.19**. The Riemann surface indicates how this multiple-valued function maps the punctured disk $0 < |z| \leq 1$ onto the real line. Namely, a vertical line passing through any point z in $0 < |z| \leq 1$ intersects this Riemann surface at infinitely many points. The heights of the points of intersection represent different choices for the argument of z. Therefore, by horizontally projecting the points of intersection onto the vertical axis, we see the infinitely many images of $G(z) = \arg(z)$.



(a) The cut disk A_0



(b) A_0 placed in xyz-space

FIGURE 2.4.18 The cut disk A_0

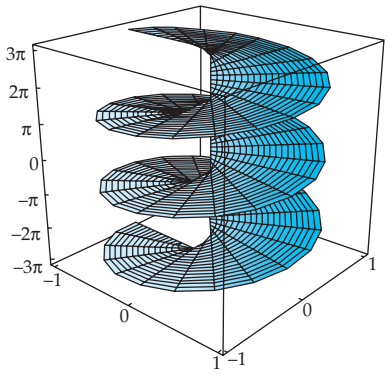


FIGURE 2.4.19 The Riemann surface for $G(z) = \arg(z)$

EXERCISES 2.4

Answers to selected odd-numbered problems begin on page ANS-8.

2.4.1 The Power Function z^n

In Problems 1–14, find the image of the given set under the mapping $w = z^2$. Represent the mapping by drawing the set and its image.

1. the ray $\arg(z) = \dfrac{\pi}{3}$

2. the ray $\arg(z) = -\dfrac{3\pi}{4}$

3. the line $x = 3$

4. the line $y = -5$

5. the line $y = -\dfrac{1}{4}$

6. the line $x = \dfrac{3}{2}$

7. the positive imaginary axis

8. the line $y = x$

9. the circular arc $|z| = \dfrac{1}{2}, 0 \le \arg(z) \le \pi$

10. the circular arc $|z| = \dfrac{4}{3}, -\dfrac{\pi}{2} \le \arg(z) \le \dfrac{\pi}{6}$

11. the triangle with vertices 0, 1, and $1 + i$

12. the triangle with vertices 0, $1 + 2i$, and $-1 + 2i$

13. the square with vertices 0, 1, $1 + i$, and i

14. the polygon with vertices 0, 1, $1 + i$, and $-1 + i$

In Problems 15–20, find the image of the given set under the given composition of a linear function with the squaring function.

15. the ray $\arg(z) = \dfrac{\pi}{3}$; $f(z) = 2z^2 + 1 - i$

16. the line segment from 0 to $-1+ i$; $f(z) = \sqrt{2}z^2 + 2 - i$

17. the line $x = 2$; $f(z) = iz^2 - 3$

18. the line $y = -3$; $f(z) = -z^2 + i$

19. the circular arc $|z| = 2, 0 \le \arg(z) \le \dfrac{\pi}{2}$; $f(z) = \frac{1}{4}e^{i\pi/4}z^2$

20. the triangle with vertices 0, 1, and $1 + i$; $f(z) = -\frac{1}{4}iz^2 + 1$

21. Find the image of the ray $\arg(z) = \pi/6$ under each of the following mappings.

(a) $f(z) = z^3$ **(b)** $f(z) = z^4$ **(c)** $f(z) = z^5$

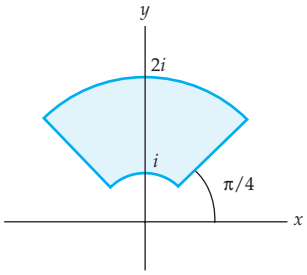


FIGURE 2.4.20 Figure for Problems 23 and 24

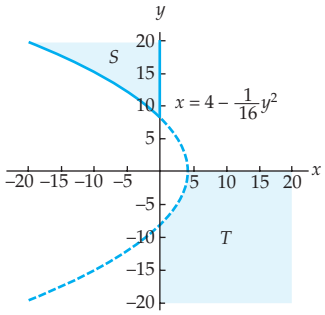


FIGURE 2.4.21 Figure for Problem 39

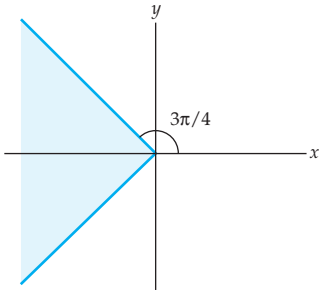


FIGURE 2.4.22 Figure for Problem 40

22. Find the image of the first quadrant of the complex plane under each of the following mappings.

(a) $f(z) = z^2$ (b) $f(z) = z^3$ (c) $f(z) = z^4$

23. Find the image of the region $1 \le |z| \le 2$, $\pi/4 \le \arg(z) \le 3\pi/4$, shown in **FIGURE 2.4.20** under each of the following mappings.

(a) $f(z) = z^2$ (b) $f(z) = z^3$ (c) $f(z) = z^4$

24. Find the image of the region shown in Figure 2.4.20 under each of the following mappings.

(a) $f(z) = 3z^2 + i$ (b) $f(z) = (i+1)z^3 + 1$ (c) $f(z) = \frac{1}{2}z^4 - i$

2.4.2 The Power Function $z^{1/n}$

In Problems 25–30, use (13) to find the value of the given principal nth root function at the given value of z.

25. $z^{1/2}$, $z = -i$

26. $z^{1/2}$, $z = 2 + i$

27. $z^{1/3}$, $z = -1$

28. $z^{1/3}$, $z = -3 + 3i$

29. $z^{1/4}$, $z = -1 + \sqrt{3}i$

30. $z^{1/5}$, $z = -4\sqrt{3} + 4i$

In Problems 31–38, find the image of the given set under the principal square root mapping $w = z^{1/2}$. Represent the mapping by drawing the set and its image.

31. the ray $\arg(z) = \dfrac{\pi}{4}$

32. the ray $\arg(z) = -\dfrac{2\pi}{3}$

33. the positive imaginary axis

34. the negative real axis

35. the arc $|z| = 9$, $-\dfrac{\pi}{2} \le \arg(z) \le \pi$

36. the arc $|z| = \dfrac{4}{7}$, $-\dfrac{\pi}{2} \le \arg(z) \le \dfrac{\pi}{4}$

37. the parabola $x = \dfrac{9}{4} - \dfrac{y^2}{9}$

38. the parabola $x = \dfrac{y^2}{10} - \dfrac{5}{2}$

39. (a) Find the image of the region S shown in **FIGURE 2.4.21** under the principal square root mapping $w = z^{1/2}$.

(b) Find the image of the region T shown in Figure 2.4.21 under the principal square root mapping $w = z^{1/2}$.

40. Find the image of the region shown in **FIGURE 2.4.22** under the principal square root function $w = z^{1/2}$. (Be careful near the negative real axis!)

Focus on Concepts

41. Use a procedure similar to that used in Example 2 to find the image of the hyperbola $xy = k$, $k \neq 0$, under $w = z^2$.

42. Use a procedure similar to that used in Example 2 to find the image of the hyperbola $x^2 - y^2 = k$, $k \neq 0$, under the mapping $w = z^2$.

43. Find two sets in the complex plane that are mapped onto the ray $\arg(w) = \pi/2$ by the function $w = z^2$.

44. Find two sets in the complex plane that are mapped onto the set bounded by the curves $u = -v$, $u = 1 - \frac{1}{4}v^2$, and the real axis $v = 0$ by the function $w = z^2$.

45. In Example 2 it was shown that the image of a vertical line $x = k$, $k \neq 0$, under $w = z^2$ is the parabola $u = k^2 - \dfrac{v^2}{4k^2}$. Use this result, your knowledge of linear mappings, and the fact that $w = -(iz)^2$ to prove that the image of a horizontal line $y = k$, $k \neq 0$, is the parabola $u = -\left(k^2 - \dfrac{v^2}{4k^2}\right)$.

46. Find three sets in the complex plane that map onto the set $\arg(w) = \pi$ under the mapping $w = z^3$.